Reinforcement learning for molecular coordinate reasoning

Results, methods, original-benchmark evaluation, and regression analysis

16 October 2026 · CoreWeave H100 training / GB200 evaluation · Marin / MarinSkyRL / Iris

Main findings

On the original PDBThink coordinate benchmark's 40 context-fitting questions, the matched rerun changes strict exact accuracy from **5/40 (12.50%)** to **11/40 (27.50%)**: **+15.00 percentage points**. Both models use the same frozen prompts, sampling settings, scoring code, and standalone serving runtime.

With the pre-existing, gold-independent terminal-box scoring correction applied to both models, accuracy changes from **8/40** to **11/40** (+7.50 points). This sensitivity analysis helps separate gains in answer formatting from gains under a less format-sensitive score.

The transfer result is encouraging but imprecise. Both paired source-group 95% intervals include zero. Box normalization reduces the net advantage from six questions to three; truncations rise from 11 to 18. This is evidence of a higher observed score on this panel, not a precise estimate of a broad reasoning improvement.

The 64-update pilot also improved a separate 1,370-task held-out test from **291 correct (21.24%) to 433 (31.61%)**, a 10.36-point gain. On the 1,003-task development validation set, native evaluation peaked at **616 correct (61.42%) at step 384**, then regressed to 334 (33.30%) at step 512. These are distinct evaluation protocols, not points on one matched before/after curve.

The regression is a real degradation in answer content as well as format. Replayed evidence is consistent with unstable RL optimization and growing disagreement between training and rollout probabilities. It does not identify a unique initiating bug. A fresh restore/parity experiment and controlled optimizer ablations remain necessary for a causal conclusion.

Scope and cutoff

Checkpoint **384** was selected using validation alone at 2026-10-16T16:52:19.400251+00:00, before original-benchmark outcomes. It is the best checkpoint available at that time, not a claim about an already completed 876-update run. Training remains active; the full-history snapshot in this report reaches update **584**, with full validation through step 576. The later 398/1,003 score (39.68%) remains below the selected peak. No original-benchmark score is used to select another checkpoint.

1. Original PDBThink benchmark

We evaluated the existing **clean-coordinate-v4.0** coordinate benchmark: 88 frozen questions spanning 19 task families. This is the original benchmark's clean, explicitly tool-free revision, not a newly sampled subset of the RL dataset. The version replaced two earlier singleton-chain inventory questions with multichain questions; the other 86 prompts were retained. Its frozen fingerprint is recorded in the reproducibility appendix.

Exactly **40/88 questions (45.45%)**, covering 18 families and 28 benchmark protein-group IDs, fit Snowball's native context. The other 48 prompts already exhaust 32,768 tokens and are excluded for both models. T01 has no eligible question. The 40-question membership and every request body were checked against the benchmark's existing Snowball plan; only model identity changes. No prompt was cropped, shortened, rewritten, or supplemented with labels.

An older Snowball benchmark run selected 25 questions by reserving 8,192 output tokens. The clean benchmark instead uses the full remaining context and admits 40. This report reproduces that later clean protocol; the 25-to-40 change is benchmark coverage, not a change in the 1,003-task RL validation set.

Score on the same 40 questions	Original	RL step 384	Change
Strict exact accuracy	5/40 (12.50%)	11/40 (27.50%)	+15.00 pp
Strict family-macro exact accuracy	13.89%	33.33%	+19.44 pp
Terminal-box-corrected exact accuracy	8/40 (20.00%)	11/40 (27.50%)	+7.50 pp
Box-corrected family macro	20.37%	33.33%	+12.96 pp
Strict format errors	18	20	Flags can overlap truncation
Native-context truncations	11	18	Zero reward

The strict comparison has **7 newly correct** answers and **1 newly wrong** answers. The approximate paired protein-group bootstrap 95% interval for the task-weighted difference is **+0.00 to +32.71 points**. Under box correction it is -4.17 to +20.51 points, with 4 gains and 1 losses.

These intervals resample the 28 benchmark protein groups with replacement 50,000 times (NumPy seed 17), retaining each model's paired responses and all questions in each group. They describe sensitivity to source composition. One sampled answer per model/question cannot quantify decoding variance; one training run cannot quantify training-seed variance. This small panel provides a direct transfer test, with limited precision for individual families.

The headline follows the clean benchmark's exact-correct criterion; partial set-overlap credit is not counted as success. The box correction is the already defined **snowball-boxed-v1** policy: unwrap only a balanced terminal \boxed{…}, normalize escaped spaces, and add FINAL: if absent. It does not repair numbers, choose among guesses using the gold answer, or relax truncation, refusal, or numerical-tolerance rules. The strict and corrected scores are always shown for both models.

Mean generated length is 3,196 tokens for the original model and 5,369 for step 384. Maximum budgets are matched; actual generated-token counts need not be equal. These counts include thinking and answer text and exclude prompt tokens.

1a. What changed on the original benchmark?

Under strict scoring, seven questions become correct and one becomes wrong. Five of the seven gains were previously format errors, one was flagged as a refusal, and one was a well-formed wrong answer. The existing box correction recovers three of those five baseline format errors as correct without RL. Consequently, the net improvement falls from six questions under strict scoring to three after applying the same normalization to both models.

After box correction, four questions become correct: three previously well-formed wrong answers and one previously refused answer. The one lost answer is a solvent-exposure question whose trained response reaches the context limit. Among the 16 questions with valid responses under both models after normalization, correct answers rise from 7 to 10. This subset is selected using the observed outputs and is descriptive, not a controlled format-only intervention.

Well-formed wrong → correct after box normalization	Original answer	RL answer	Gold
G01 / fb_7t9r: atom distance	8.22 Å	8.81 Å	8.812 Å; tolerance 0.02
G03 / fb_7t9r: nearest residue	A:C29	A:K38	A:K38
S09 / fb_8aga: side-chain rotamer	g+	t	t

The G01 and G03 gains concern the same protein group, fb_7t9r. They are correlated evidence, not two independent protein-level replications. The benchmark's small, uneven family denominators also make its family-macro gain sensitive to single questions: G01 and G03 each contain only one eligible question.

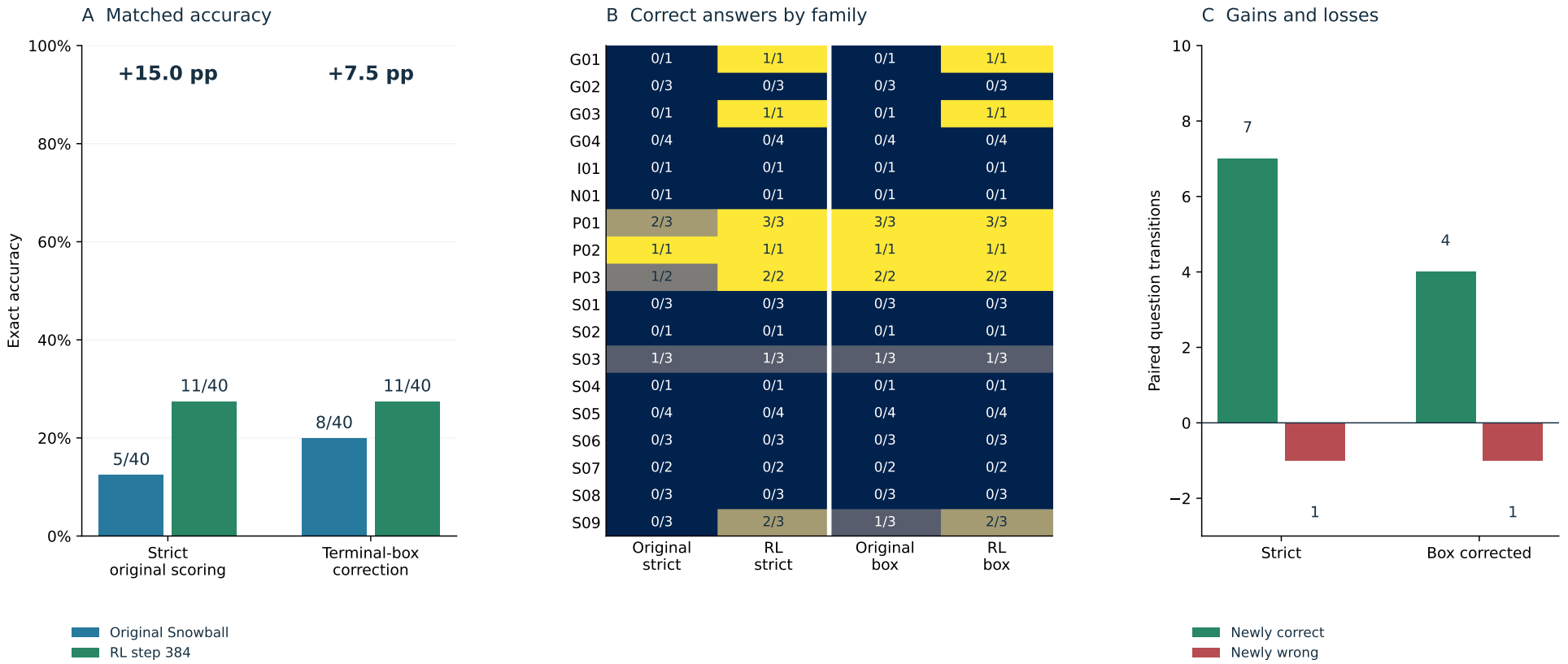
Correct final answers do not certify correct reasoning or clean surrounding output. The three responses above include invented CID/RID metadata and malformed surrounding JSON. The G03 explanation describes a Cα-based comparison even though the question asks for minimum heavy-atom distance. The unchanged original parser can extract the correct FINAL value, so these count as benchmark successes. They are not evidence that every stated intermediate computation or invented metadata field is valid. Here, "strict" means the original scoring policy without the additional terminal-box correction; it does not mean a new, stricter whole-response validator.

Trained exact accuracy is still zero in 11 of the 18 eligible families. Overall strict format-error flags increase from 18 to 20; refusal flags fall from 8 to 0; truncations increase from 11 to 18. The flags can overlap. Mean generated length rises from 3,196 to 5,369 tokens, about 68% more output tokens under matched maximum budgets. The useful answer improvements therefore coexist with substantial failure rates and higher realized generation cost.

2. Transfer to the original benchmark

Does the RL checkpoint improve the original PDBThink benchmark?

Same 40 context-fitting questions · same GB200 inference runtime · checkpoint selected on validation alone



Paired protein-group bootstrap 95% interval for improvement: strict +0.0 to +32.7 pp; box corrected -4.2 to +20.5 pp.
40/88 questions fit; 48 are context-excluded. One response per model/question at temperature 0.6. Intervals exclude training-seed and decoding uncertainty.

Matched inference on 40 frozen questions; both models use temperature 0.6 and the full remaining native context. Strict scoring is primary; terminal-box normalization is a fixed sensitivity analysis. Family denominators are small and unequal. The footer intervals resample benchmark protein groups and exclude training/decoding uncertainty.

3. Exact benchmark inference method

Setting	Value / rationale
Baseline	Original Snowball Step38, revision cfc1d845dae89b067cdc7250d0164abefa5a69cf
RL model	One validation-selected native checkpoint: global_step_384; native Megatron-to-HF export
Serving	Same Marin standalone vLLM configuration for both; 4 CoreWeave GB200 per server, BF16, TP1 / DP4 with expert parallelism
Sampling	One response per question; temperature 0.6, top-p 0.95, top-k 20; seed 20260929; thinking enabled
Stop tokens	128001 and 128009, as in the frozen original Snowball benchmark requests
Context / output budget	32,768 native tokens; max_tokens = 32768 − exact chat-template prompt tokens; remaining budgets range from 1,191 to 20,136
Tool policy	tools=null and tool_choice=none on every request; raw responses audited for tool events; no tool execution
Concurrency	4 requests in both new evaluations; historical deployment used 2. Historical scores are not used as the paired baseline.
Verifier	Existing benchmark scorer and question parameters; source file hashes frozen before inference; strict and pre-existing boxed policy both retained
Failures / resumption	Only transient transport/service errors retried, at most three attempts with identical request; model errors never retried. Each completed raw response saved atomically.

The runner checks served prompt-token counts against locally computed counts and rejects an unexpected output-budget clamp. The checkpoint export is checked against its native publication manifest, including every file’s size and metadata hashes; tokenizer and chat template must match the frozen baseline. Native publisher weight hashes are preserved, but the large weight shards are not downloaded a second time solely for checksum replay.

As a scorer cross-check, all 40 historical original-benchmark responses were replayed under both scoring policies: 80 decisions, zero disagreements with the published clean report. That historical deployment scored 5/40 strictly and 10/40 after terminal-box normalization. These historical values are context only; the RL improvement is computed against the fresh baseline on the same GB200 serving stack as the trained model.

The original benchmark uses sampled decoding (temperature 0.6), while RL development validation uses greedy decoding (temperature 0). Its full-context membership also differs from the RL cohort’s 8,192-token reserve rule. Therefore the original benchmark should be interpreted as its own matched endpoint comparison. Absolute scores need not match either native RL validation or the pilot’s standalone held-out evaluation.

4. Model, data, and split construction

The starting model is **open-athena/Snowball-67B-A2B-5.7T-Mixed-RLVR-Step38**, a mixture-of-experts model named for approximately 67 billion total and 2 billion active parameters. “Step38” identifies the upstream checkpoint; step 0 in this report is the start of our additional RL, not an untrained language model. We use the pinned native tokenizer and thinking chat template throughout.

Training uses **open-athena/pdbthink-coordinate-tasks v1.3.0**, a 100,000-task release generated from 2,671 source entries and 1,867 source groups. Published splits are retained. The release audit reports zero overlap with configured benchmark sources by entry ID, exact protein sequence, and RCSB 30% sequence-cluster membership, and zero duplicate prompts or semantic tasks. These safeguards concern this training dataset; they do not prove absence from the foundation model’s pretraining.

For this experiment, every selected prompt is retokenized with Snowball’s exact native template, including thinking-mode setup. Eligibility requires at most **24,576 input tokens**, leaving at least 8,192 of the 32,768-token context for an answer. This produces 28,045 training tasks, 1,003 validation tasks, and 1,370 test tasks. The filter changes family and class frequencies substantially; these are context-eligible subsets, not representative samples of every released task.

Cohort	Tasks	Groups	Use
Training	28,045	438	Sampled prompts used for policy updates
Full validation	1,003	20	Development and checkpoint selection
Pilot monitor	107	20	Fixed subset of validation, at most 8 per family
Held-out test	1,370	29	Paired pilot test; final selected-model test is a separate follow-up
Original benchmark	40 / 88	28 eligible benchmark IDs	External frozen transfer panel; separate context rule

Release source groups connect entries sharing exact protein sequences, RCSB sequence clusters, or two-state partners; entire connected groups remain in one split. Thus many task rows are correlated observations from the same structure or related source. The benchmark’s protein_group_id units are its own grouping identifiers, not the 20 validation-source groups.

Prompts contain sanitized, rigidly rotated and translated coordinates, with source-identifying annotations removed. Molecular questions are answered from the displayed coordinates under deterministic geometric definitions. Gold answers are verifier metadata and are never included in model-visible messages. **No GLM responses, other teacher traces, supervised trace fitting, or distillation were used.**

5. Cohort composition by task family

Family / task	Train	Validation	Test	Original eligible
G01 Atom distance	3015	110	136	1
G02 Nearest nonbonded atom	3335	120	151	3
G03 Nearest residue	2892	108	130	1
G04 Worst steric clash	58	3	2	4
I01 Interface contacts	12	0	0	1
N01 Shared contact residue	1945	72	126	1
P01 Chain inventory	918	32	40	3
P02 Residue count	434	23	28	1
P03 Atom coordinates	3339	118	152	2
S01 Salt bridge	1181	50	82	3
S02 Phosphorylated residue	436	9	8	1
S03 Solvent exposure	3334	119	152	3
S04 Local secondary structure	3124	111	138	1
S05 Fold class	314	12	23	4
S06 Ligand contacts	53	0	0	3
S07 Metal coordination	92	0	3	2
S08 Disulfide partner	278	0	50	3
S09 Side-chain rotamer	3285	116	149	3
T01 Two-state comparison	0	0	0	0

Validation covers 14 families. I01, S06, S07, S08, and T01 are absent, so the validation peak says nothing directly about those tasks. The original benchmark adds eligible I01/S06/S07/S08 questions, but still no T01. Several families have only one to four original-benchmark questions; a single success can shift their family accuracy dramatically.

The 107-question pilot monitor uses exactly eight tasks per validation family except G04, which has only three. It is a fixed subset, not a new random sample every evaluation. The full validation set contains the monitor tasks and never changes membership during the recorded long run.

6. How the RL updates were computed

We used **group-relative policy optimization (GRPO)** through Marin's pinned MarinSkyRL launcher, with Megatron training and vLLM response generation. Each update samples 32 training prompts and generates four independent answers per prompt, for 128 responses. Rollouts use temperature 1.0, top-p 1.0, top-k −1 (disabled), min-p 0, repetition penalty 1, thinking mode, and an 8,192-token answer cap. There is one optimizer pass over each batch.

The frozen native PDBThink verifier reads the model's original response text and the task's hidden gold metadata. Reward R is 1 only for a correctly formatted, exactly correct answer under the task's numerical or set-matching rules; it is otherwise 0. For example, numeric correctness uses the task tolerance rather than literal string identity. Truncation, refusal, tool calls, and oversized outputs receive zero reward. Partial-credit diagnostics do not enter the reward.

Relative advantages

Within each prompt's group of four responses, MarinSkyRL computes the mean reward and sample standard deviation. Each response receives advantage $A_i = (R_i - \text{mean}(R)) / (\text{std}(R) + 10^{-6})$. That scalar is broadcast over its generated-token mask. Groups whose four rewards are identical have zero centered advantage and provide no outcome-reward gradient. No additional whole-batch advantage normalization is enabled, and no learned critic is used.

Policy objective and optimizer

The configured loss is regular PPO with token-mean reduction: minimize the negative mean of $\min(r_t A_t, \text{clip}(r_t, 0.8, 1.2)A_t)$ over response tokens, where r_t is the exponential difference between the current action log probability and the stored old action log probability. Prompt tokens are masked from the loss. Thinking tokens have weight 1.0; longer responses contribute more token positions under token-mean reduction.

Optimization setting	Resolved value
AdamW learning rate	1×10^{-6}; constant schedule; 0 warmup steps
Adam coefficients / decay	β_1=0.9, β_2=0.999; weight decay 0.01
Gradient norm limit	1.0; diagnostic raw norm is measured before clipping
Batch / microbatch	32 prompt groups × 4 answers; per-GPU microbatch 1; one optimizer pass
PPO ratio clip interval	[0.8, 1.2]; see the effective-clipping caveat below
Reference KL / entropy / off-policy correction	All disabled; unused positive default coefficients do not make these losses active
Random seed	17 for the training recipe
Training extent	Pilot 64 updates; continuation targets 876 cumulative updates

In this pinned implementation, old action probabilities are recomputed by the learner before the update. With one pass and no intervening parameter update, the recorded ratio is exactly one and PPO clipping is inactive. This is expected behavior of this recipe, not evidence that the clip function is broken. It also means that this PPO term does not bound the actual Adam parameter step or the mismatch to the rollout policy. Gradient clipping remains a separate active mechanism.

7. Infrastructure and checkpoint lineage

All training used CoreWeave H100 GPUs managed by Marin's Iris infrastructure. The 64-update pilot and long continuation each allocate **48 GPUs**: four 8-GPU policy workers and two 8-GPU rollout workers. Policy geometry is tensor parallelism 1, pipeline parallelism 2, and expert parallelism 16 (TP1/PP2/EP16). Each rollout worker uses data parallelism 8 and expert parallelism 8. The earlier two-update smoke used only one rollout worker (40 GPUs total).

Policy workers use BF16, gradient checkpointing, per-GPU microbatch size one, 256-token chunks for log-probability calculation, and optimizer-state offload during rollout. Gradient reduction is not forced to FP32. MoE router replay, train/eval parity checks, and data-parallel weight-consistency checks are disabled in the resolved recipe. Each training node requests 64 CPU cores, 1,800 GB host RAM, and 1,000 GB disk.

LoRA adapters are disabled (rank 0); this is policy-weight training, with the Grug query-bias update mode frozen as specified in the resolved configuration. Checkpoint conversion and both original-benchmark inference jobs use CoreWeave GB200 capacity because the H100 pool was saturated. The first H100 baseline attempt was preempted before it returned answers. This hardware change affects neither the training lineage nor which checkpoint was selected.

Stage	Updates / outcome	Meaning for lineage
Pilot, 7–8 October	1–64 completed	2,048 prompts at most; exported step 64
First long attempt, 8–10 October	65–375; failed on required rollout archive upload	Last complete checkpoint 320; updates 321–375 discarded
Resumed run, 15 October onward	Restored policy, optimizer, scheduler, and dataloader at 320	Repeats updates 321 onward; target 876
Recovery change	Archive-upload timeout 120 → 600 seconds	No change to model, data, optimization, or evaluation recipe
Selected checkpoint	384; 616/1,003 native validation	Selection frozen before original-benchmark inference
Current report snapshot	Through update 584	Training still active; not a completed 876-update report

The first attempt stopped on an infrastructure publication timeout, not on a detected model-quality threshold. Resumption from 320 drops the unsaved 55-update tail. The full-history plots show that discarded tail separately and do not treat it as part of the selected model's lineage. W&B; did not retain the pilot's final optimizer-diagnostic row at 64; training completion is independently verified, but no missing metric is fabricated or interpolated.

Every 64-step full-validation checkpoint in the long run is banked by a pending native HF export request, which protects it from ordinary rolling checkpoint retention. Selection maximizes full-validation task-weighted accuracy; ties use family-macro accuracy, then the earlier step. Export restores the native 32-rank policy geometry on 32 CoreWeave GB200 GPUs (8 nodes × 4), with the same TP1/PP2/EP16 layout and converts it to a complete Hugging Face model. The original-benchmark servers use 4 GB200s each and release their owned resources after evaluation.

At the 876-update target, 876 × 32 = 28,032 prompt slots are processed in the retained one-pass schedule; the final incomplete batch of 13 training tasks is omitted. The pilot is the initial segment of that lineage, not an extra full epoch. Repeated updates after the failed attempt consume additional compute but are not extra retained updates.

8. Validation sets and protocol changes

Measurement / phase	Tasks	When	Sampling / budget
Pilot native monitor	107 fixed validation tasks	Before training; steps 8,16,...,64	Temperature 0; one answer; full remaining native context
Standalone full validation	Same 1,003 frozen tasks	Original model (0) and pilot checkpoint (64)	Temperature 0; one answer; full remaining context
First long-run native full validation	Same 1,003 frozen tasks	64 before continuation; 128,192,256,320	Temperature 0; one answer; native training engine
Resumed native full validation	Same 1,003 frozen tasks	320 before updates; 384,448,512,576 in this report	Unchanged task membership and native evaluation recipe
Pilot held-out test	1,370 tasks from 29 groups	Paired original and preselected pilot step 64	Standalone; temperature 0; full remaining context
Original benchmark	40 eligible of 88	Original and validation-selected step 384	Original protocol: temperature 0.6; one answer; full remaining context

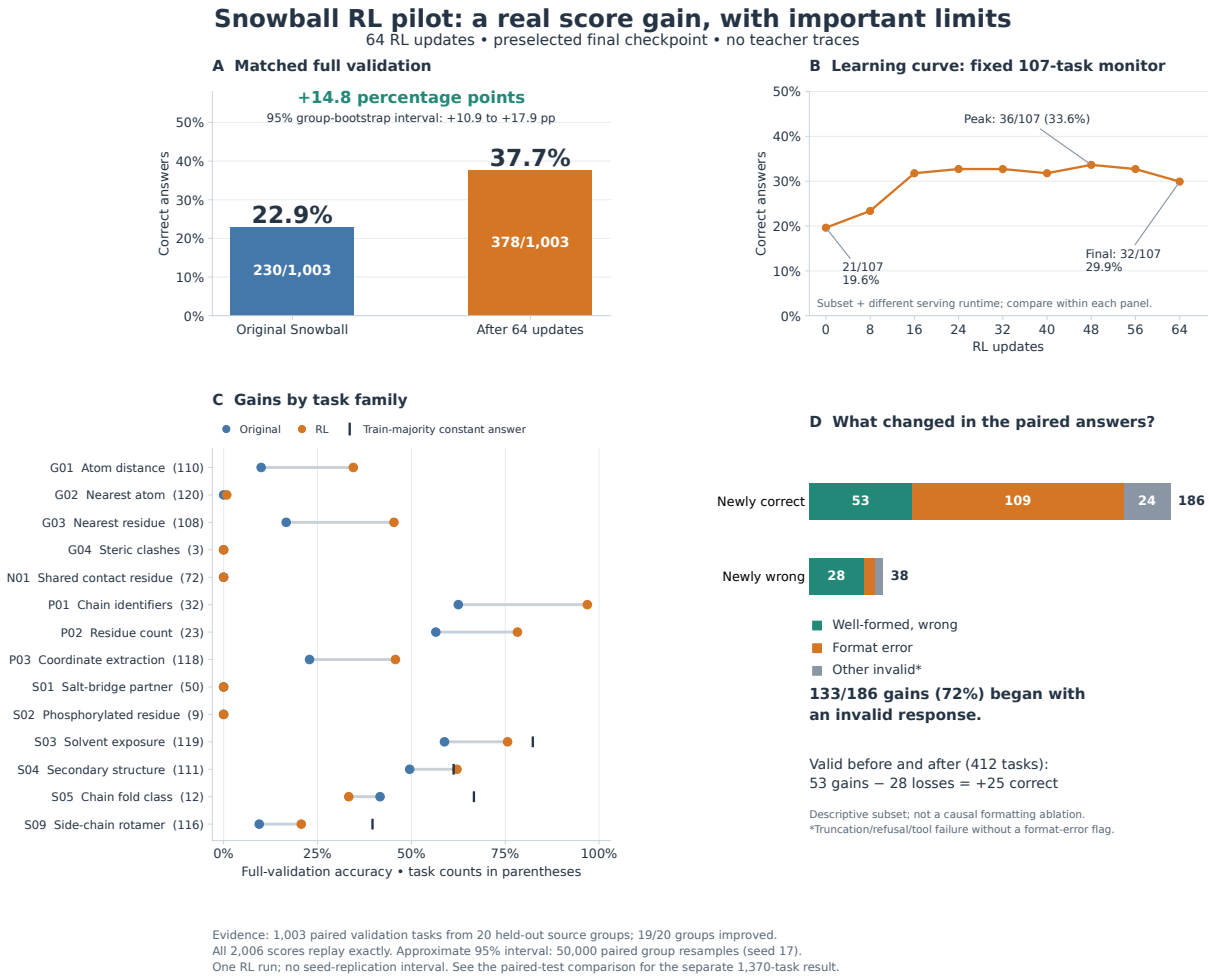
The principal cohort change is at the pilot-to-long-run boundary: native monitoring expands from 107 to all 1,003 validation tasks. The standalone 1,003-task baseline/pilot comparison is a separate measurement. From the first long-run evaluation at step 64 onward, the native validation membership does not change. The step-320 restart repeats the same set; 576/1,003 before interruption versus 580/1,003 immediately after restore differs by four answers before any new RL update. Those four answers are not evidence of learning.

Standalone serving and the native training evaluator use different execution paths and vLLM fork revisions. At step 64, standalone full validation is 378/1,003 (37.69%) while native full validation is 401/1,003 (39.98%). The mean output length also differs materially. Consequently, subtracting the standalone step-0 score of 22.93% from the native step-384 score of 61.42% is not a controlled estimate of RL improvement.

All evaluation budgets use 32,768 minus exact native prompt length. The 8,192-token reserve determines RL cohort membership, and the 8,192-token training cap bounds rollout generation; neither is a fixed validation output cap. The original benchmark uses a separate full-context eligibility rule, so it includes prompts with less than 8,192 tokens remaining.

The training monitor, full validation, and test each have a role: monitor and full validation are repeatedly observed development evidence; the final test is not used for checkpoint selection. The original benchmark is a separate transfer evaluation requested after observing the learning curve. Results have been inspected for the pilot test already, so it should not be described as forever untouched or as an independent model-selection set.

9. What the pilot established



Evidence: 1,003 paired validation tasks from 20 held-out source groups; 19/20 groups improved.
All 2,006 scores replay exactly. Approximate 95% interval: 50,000 paired group resamples (seed 17).
One RL run; no seed-replication interval. See the paired-test comparison for the separate 1,370-task result.

Pilot evidence snapshot: full standalone validation compares the identical 1,003 prompts at step 0 and step 64. Group-bootstrap uncertainty pertains to source composition. The completed 1,370-task paired test is reported in the following section.

10. Pilot results and their interpretation

Paired measurement	Original	Pilot step 64	Difference
Full validation (1,003)	230 / 1,003; 22.93%	378 / 1,003; 37.69%	+14.76 pp
Validation family macro	23.44%	35.25%	+11.81 pp
Held-out test (1,370)	291 / 1,370; 21.24%	433 / 1,370; 31.61%	+10.36 pp
Native monitor (107)	21 / 107; 19.63%	32 / 107; 29.91%	+10.28 pp

The validation comparison has 186 newly correct answers and 38 newly wrong answers; the test has 192 gains and 50 losses. Nineteen of the 20 validation source groups improve. The paired source-group bootstrap interval for validation improvement is +10.86 to +17.94 percentage points (50,000 resamples, seed 17). All 2,006 validation responses replay exactly through the native verifier, and 336 G01/G03/P03 coordinate-derived gold answers were independently checked without disagreement.

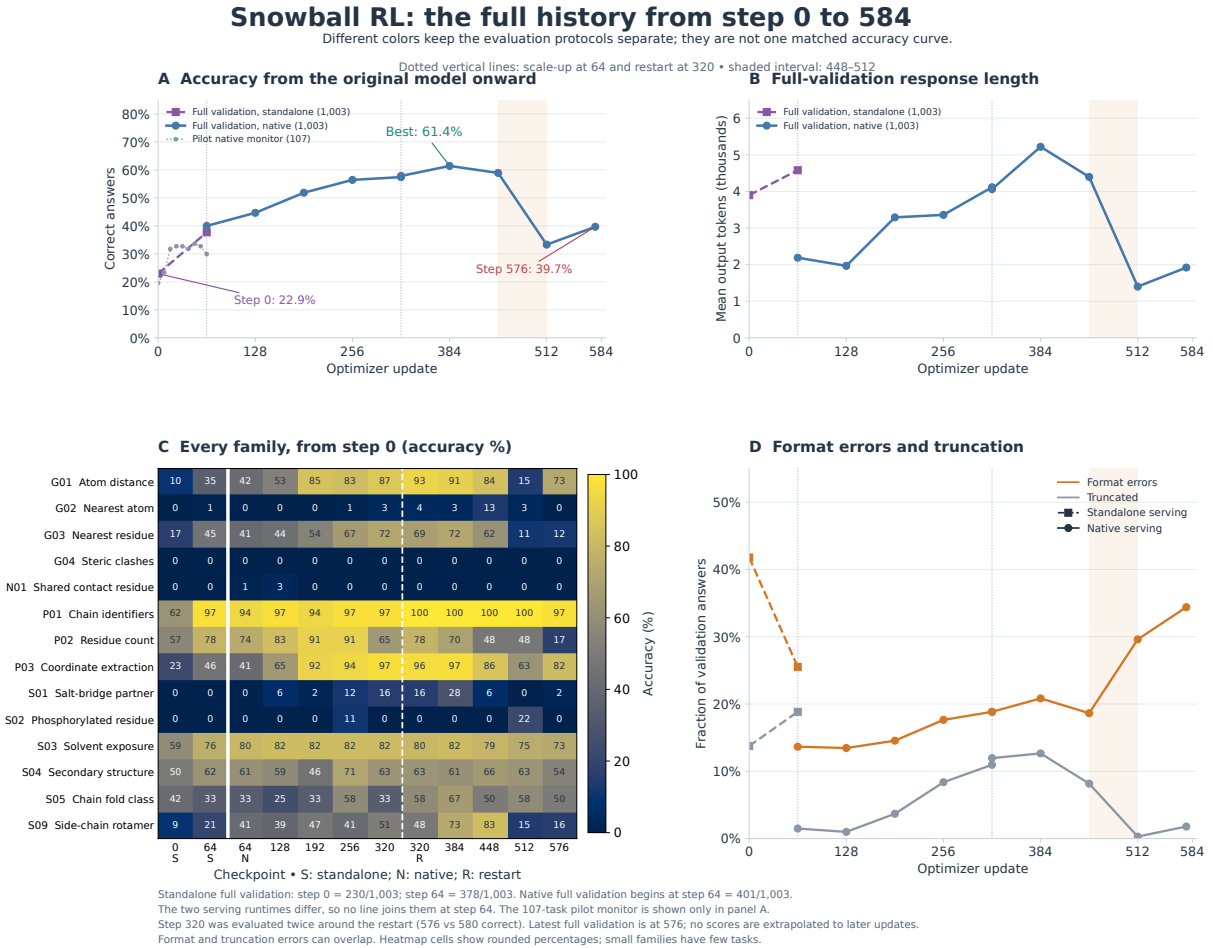
However, **133 of the 186 validation gains (71.5%)** were previously invalid outputs: 109 format errors and 24 other invalid responses. Only 53 gains came from previously well-formed wrong answers. Among 412 tasks valid under both models, correct answers rise from 220 to 245. That output-selected subset is descriptive, not a causal estimate of what would happen if formatting alone were repaired.

Categorical family	Fixed training-majority answer	Control	Original	Pilot
S03 exposure	solvent-exposed	98/119	70/119	90/119
S04 secondary structure	coil	68/111	55/111	69/111
S05 fold class	predominantly alpha helical	8/12	5/12	4/12
S09 rotamer	g−	46/116	11/116	24/116

The constant labels are chosen from training data, not validation labels. The pilot predicts coil on 110/111 S04 tasks; it exceeds that trivial control by only one correct answer. Three other categorical families remain below their constant controls. G04, N01, S01 and S02 remain at zero, and G02 is only 1/120. These results support useful benchmark adaptation, with narrower evidence of improved geometric computation or general structural understanding.

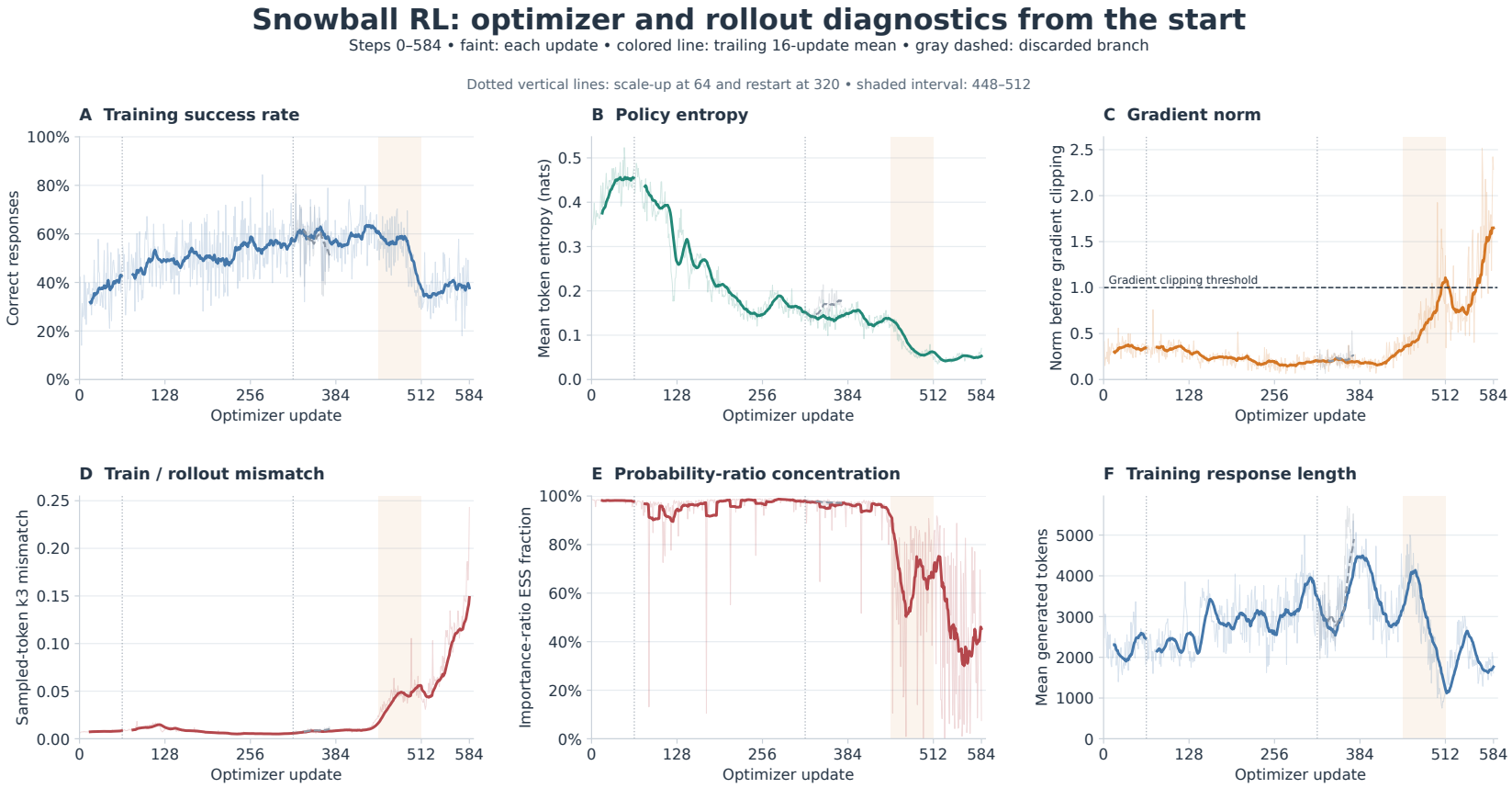
Pilot validation format errors decline from 419 to 256, while truncations rise from 138 to 189 and mean generated length rises from about 3,898 to 4,580 tokens. The same maximum token allowance does not imply equal actual inference compute. The improvement is real under the chosen evaluation protocol, but it is not a claim of equal-compute gains or broad mastery of every family.

11. Full recorded learning curve from step 0



Step 0 means the starting Snowball model. Gray: fixed 107-task native pilot monitor. Purple/dashed: paired 1,003-task standalone validation. Blue/solid: 1,003-task native training validation, with a separate segment at restart. Heatmap labels S=standalone, N=native, R=restart. Every plotted time axis begins at zero; missing optimizer observations are not invented.

12. Full recorded optimizer diagnostics



Retained lineage: pilot 1–64 → first continuation 65–320 → resumed continuation 321 onward. The failed attempt's updates 321–375 were discarded and repeated.
Update diagnostics first exist at step 1. W&B is missing pilot update 64; gaps are left unfilled. The plotted history contains 583 retained update records.
Mismatch compares Megatron training with vLLM rollout probabilities, not the original model. ESS describes token probability ratios, not independent tasks.

Faint traces are raw per-update values; the emphasized curve is a trailing 16-update mean. The discarded first-attempt tail at steps 321–375 is shown separately in gray. Vertical dotted lines mark the pilot/continuation boundary at 64 and restore at 320. Shading highlights 448–512, the interval bracketing the validation collapse; it is not an estimated exact change point. Definitions follow in the glossary.

13. The regression is an answer-quality loss

Native validation	Correct / 1,003	Format errors	Truncated	Mean output tokens
320 after restore	580 (57.83%)	189	120	4,063
384	616 (61.42%)	209	127	5,220
448	591 (58.92%)	187	82	4,397
512	334 (33.30%)	297	3	1,400
576	398 (39.68%)	345	18	1,920

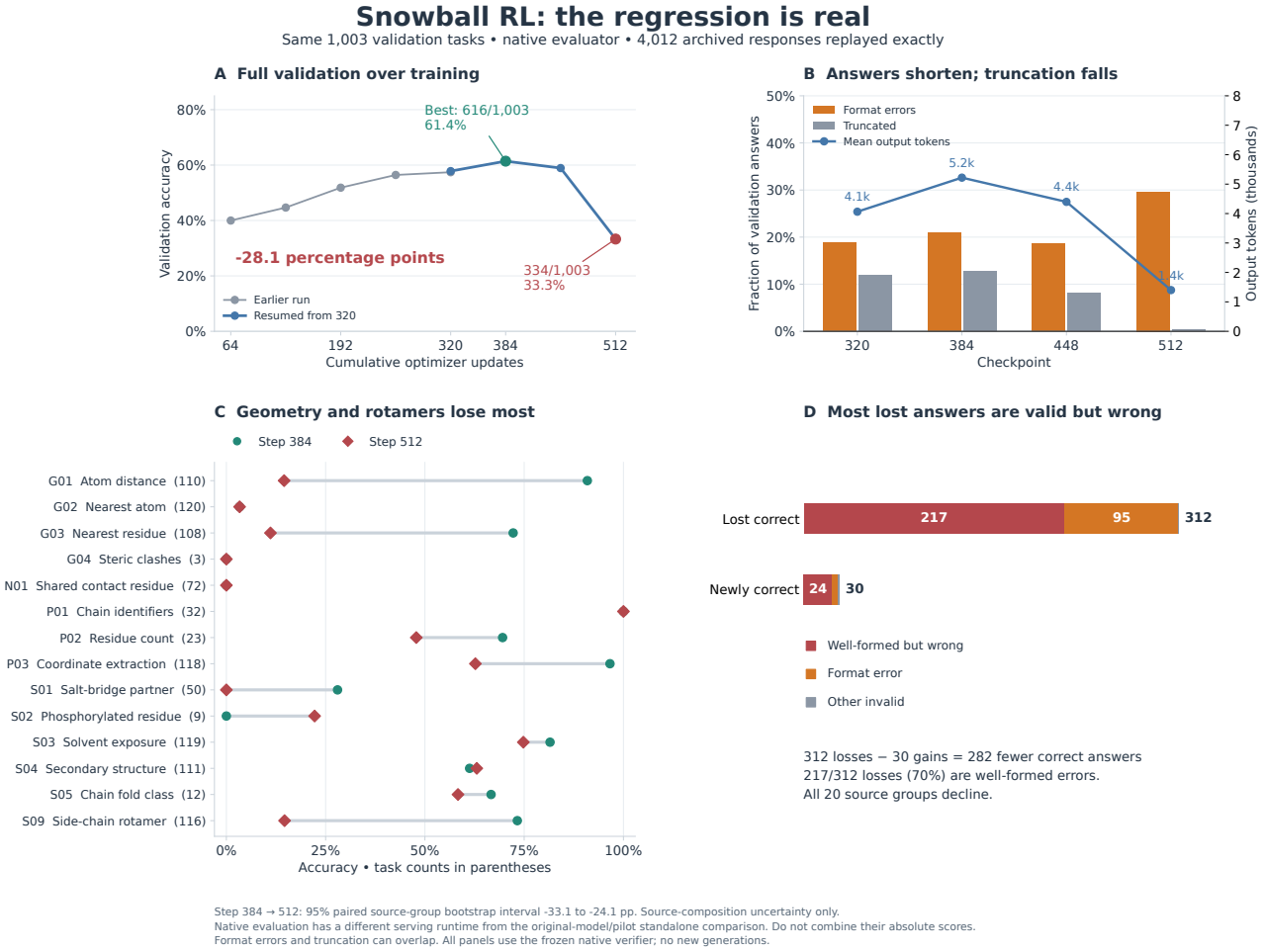
From step 384 to 512, accuracy falls **28.12 percentage points**. There are 312 losses and 30 gains, with 304 tasks correct under both checkpoints and 385 wrong under both. All 20 validation source groups decline. A paired source-group bootstrap interval for the change is −33.05 to −24.07 points. This is much larger than the observed four-answer restart variation at step 320, though no single interval captures all serving or training randomness.

Of the 312 lost correct answers, **217 become well-formed wrong answers** and 95 become format errors. Thus formatting cannot explain the whole collapse. Truncations become rare, while answers become much shorter. This is not a failure driven by running out of answer budget.

Family	Step 384	Step 512	Observed failure
G01 atom distance	100/110	16/110	Large numerical reasoning loss; no format errors at 512
G03 nearest residue	78/108	12/108	Candidate selection degrades
P03 coordinates	114/118	74/118	Previously reliable extraction degrades
S09 rotamer	85/116	17/116	All 54 valid outputs choose t; 62 format-invalid

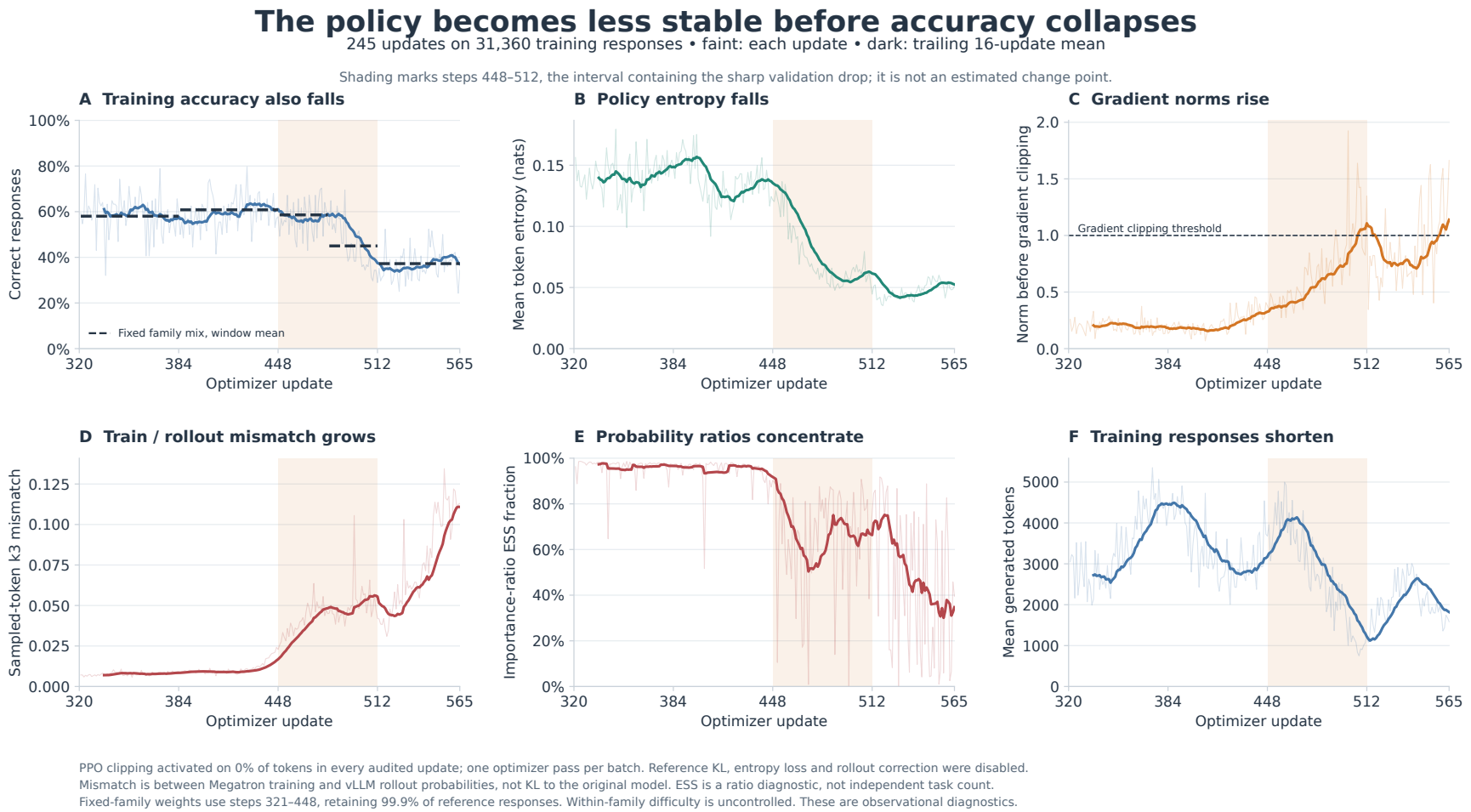
A deterministic example, selected as the first lost G01 task by ID, illustrates the content change. Coordinates are (−0.792, −17.922, −10.053) and (2.386, 0.974, −7.489); gold distance is 19.332 Å. Step 384 computes the differences (3.178, 18.896, 2.564) and answers 19.33. Step 512 extracts the coordinates but mixes axes in a two-dimensional expression and answers 17.32. This example illustrates an audited transition; it is not an estimate of how frequently that particular error occurs.

14. Detailed regression evidence



Audit window: native full validation at 320/384/448/512, with paired transitions and source-group uncertainty for 384→512. The original saved audit covers those checkpoints; the full-history figure separately includes the later 576 evaluation. Each comparison uses the same frozen 1,003 tasks.

15. Diagnostics around the collapse



The audit's response replay and fixed-family-weight analysis cover resumed updates 321–565. Family standardization holds the reference family mixture fixed; it cannot remove changes in task difficulty within a family. All measured training and validation rewards were replayed from original responses. See the glossary for k3 mismatch, ESS, gradient norm, and entropy.

16. What caused the regression?

The strongest supported conclusion is **policy degradation during continued RL, associated with a loss of optimization stability**. It is supported by a fixed validation set, paired raw-answer losses, falling sampled training reward after family adjustment, increasing gradient norms, and growing training-versus-rollout probability mismatch. The evidence is observational: it identifies a credible mechanism and rules out several simpler explanations, but does not establish a unique trigger.

Checks that passed

All 4,012 native validation responses from checkpoints 320/384/448/512 reproduce their recorded rewards and diagnostics when replayed using original API response content. All 31,360 sampled training responses in 7,840 four-response prompts from updates 321–565 also replay exactly. Prompt hashes, native tokenization, family/source metadata, and gold metadata agree. The audit checks 7,844 native archive hashes and independently recomputes 336 coordinate-derived gold answers. It finds no reward mismatch, stale-policy record, token-alignment failure, skipped optimizer step, or upload timeout in that audited interval.

Holding the family mixture fixed to updates 321–448 gives sampled success rates of 58.02% at 321–384, 60.81% at 385–448, 58.58% at 449–480, 44.99% at 481–512, and 37.24% at 513–565. Common families cover 99.85% of reference prompts; I01/S06 are excluded because they are absent from at least one window. This reduces the family-composition explanation but does not control within-family difficulty.

A concrete stability weakness

The learner recomputes old action log probabilities with its current weights, and there is one optimizer pass. Every one of the 245 audited updates has PPO clip fraction 0 and exact-unit ratio fraction 1. The configured PPO ratio term therefore does not constrain these updates against the actual rollout probabilities. There is also no reference-KL loss, no entropy bonus, and no off-policy correction. The active gradient norm limit is a separate safeguard, not a replacement for those constraints.

Comparing updates 321–384 with 545–565, mean token entropy falls from 0.14138 to 0.05278 nats, raw gradient norm rises from 0.19749 to 1.12197, k3 training/rollout mismatch rises from 0.007837 to 0.106081, and ESS fraction falls from 0.96285 to 0.36206. The wider history shows entropy falling during successful learning too, so entropy decline alone is not specific evidence of collapse. The later mismatch and gradient-norm changes are more diagnostic.

What remains unresolved

Zero nominal policy-version staleness does not prove numerical parity between Megatron and vLLM. Possible contributors include MoE routing, precision/backend differences, or weight synchronization; the disabled parity/router-replay checks leave these unresolved. A fresh standalone evaluation of the selected checkpoint tests transfer after restore, but does not by itself isolate the training/rollout mismatch or explain checkpoint 512.

The next causal experiment should restore 384 and 512 in fresh workers and check weight and log-probability parity, then compare matched continuations from 384 under the current recipe and behavior-referenced clipping, with learning-rate changes tested separately. These ablations have not been run. It would be premature to claim that a smaller learning rate, a specific clipping option, or a synchronization fix has already solved the regression.

17. Reading the plots: accuracy and cohorts

Term	Precise meaning
Step / optimizer update	One policy update after a batch of 32 prompts × 4 sampled responses. Step 0 is the supplied Snowball checkpoint before this experiment's RL.
Success rate / accuracy / reward	Fraction of responses with binary reward 1: valid and fully correct under the native verifier. A numeric answer may match within a specified tolerance. Partial credit is excluded.
Training reward	Mean reward on that update's sampled training responses at temperature 1 and an 8,192-token cap. Prompts change with batches; this is not held-out accuracy.
Pass@1 / pass@4	Pass@1 is success for one response. Training pass@4 is the fraction of prompt groups with at least one success among four samples; it is not the mean success over all four.
Task-weighted / micro accuracy	Total correct divided by total questions. Common families and larger source groups contribute more tasks.
Family-macro accuracy	Compute accuracy within each included family, then average those family accuracies equally. Excluded/absent families are not silently assigned zero.
Percentage points (pp)	Difference between percentages: 37.69% −22.93%=14.76 points; not a 14.76% relative increase.
Validation / monitor / test	Validation guides development and checkpoint selection. Monitor is a fixed 107-task validation subset. The 1,370-task test is a separate grouped split. The original 88-question benchmark is another dataset.
Source group	A unit of related structures/tasks kept together by sequence/cluster/two-state relationships in the RL release; benchmark protein-group IDs use the original benchmark grouping.
Bootstrap interval	A percentile interval from resampling entire paired source groups, retaining their tasks. It quantifies source-composition sensitivity, not variation across training seeds or inference stacks.
Gains / losses	Gains are wrong→correct paired transitions; losses are correct→wrong. Correct under both and wrong under both complete the transition table.
Fixed family mix	Reweights per-family training success to a common reference mixture. It adjusts family proportions, not task difficulty within a family.
S / N / R on heatmap	S=standalone serving; N=native training evaluator; R=the repeated native evaluation immediately after restart.
Dotted lines / shaded band	Dotted lines mark 64 and 320 phase boundaries; 448–512 shading brackets the observed validation drop, not its exact onset.

18. Reading the plots: optimizer and output

Term	Precise meaning and limitation
Token / output length	A tokenizer unit. Output-token counts include generated thinking and final-answer text, not input coordinates. The model's context limit covers input plus output.
Format error	The scorer cannot parse an answer under the requested schema. A well-formed wrong answer parses but fails correctness. A format error and truncation may overlap.
Truncation	Generation reaches its allowed token budget. It receives zero reward even if a partial answer looks correct. A shorter average response is not automatically an improvement.
Policy entropy (nats)	Average $-\Sigma\ p \log(p)$ of the learner's next-token distribution over scored response positions. Natural logarithms give nats. Lower entropy means a more concentrated distribution, not greater correctness.
Raw gradient norm	Euclidean norm of the policy gradient before clipping. The configured limit is 1.0. This is not the magnitude of the Adam parameter update, which also depends on optimizer state.
PPO clipping / clip fraction	Clips the probability-ratio surrogate when it changes the objective in the relevant advantage direction. Different from gradient clipping. In this one-pass recipe the recorded ratios equal 1, so clip fraction is 0.
Training/rollout ratio r	$r=\exp(\log p_{training} - \log p_{rollout})$ for the sampled token. It compares learner and generator probabilities of the same observed token; nominal policy versions can match while numerical probabilities differ.
k3 mismatch	Mean $r-1-\log(r)$, with the pinned implementation's numerical safeguards, over valid sampled tokens. A nonnegative KL-style mismatch diagnostic. It is not a percentage, an accuracy, or KL distance from the original Snowball model.
ESS fraction	$(\Sigma r)^2 / (N\Sigma r^2)$, the importance-weight effective-sample-size ratio. Near 1 means weights are nearly uniform, not necessarily exactly 1. It is not the number of independent biological tasks.
Token alignment / TIS diagnostics	Alignment diagnostics confirm that learner and generator logs refer to matched tokens. Their presence under a generate/tis metric name does not mean importance-sampling correction is enabled.
Rolling mean	A trailing 16-update average used only to make trends readable; faint raw traces retain per-update variation. It is not a confidence interval.
BF16 / TP / PP / EP / DP	BF16 is the 16-bit floating-point format. Tensor, pipeline, expert, and data parallelism respectively partition tensor operations, layers, MoE experts, and data processing across devices.
GRPO / PPO / KL	Group-relative policy optimization; proximal policy optimization; Kullback–Leibler divergence. The active recipe uses GRPO advantages and regular PPO loss, with reference-model KL disabled.

19. Molecular task glossary

Family	What the question asks
P01 / P02 / P03	List polymer chain IDs; count residues in a named chain; extract a named atom's displayed (x,y,z) coordinates.
G01	Euclidean distance between two named atoms, in ångströms (Å; 10^{-10} m).
G02	Nearest eligible heavy atom, excluding covalent neighbors one or two bonds away. Heavy atoms exclude hydrogen.
G03	Closest candidate residue using minimum heavy-atom distance under the task definition; not merely Cα-to-Cα distance.
G04	Identify the residue pair containing the most severe atom-level steric clash. Overlap is $r_1+r_2-0.40-\text{distance}$, using specified Bondi radii and exclusions for bonded neighbors, metals, and SG–SG pairs.
N01	Identify the unique residue contacting both named anchor residues. Contact means at least one heavy-atom pair lies within the specified cutoff (4 Å in the audited prompts).
I01	Residue contacts at a specified chain interface, using the frozen distance and counting rules.
S01	Salt-bridge partner: relevant Lys/Arg nitrogen to Asp/Glu oxygen distance at most 4 Å, according to the prompt's named atoms.
S02	Identify a phosphorylated serine, threonine, or tyrosine in the supplied structure.
S03	Classify solvent exposure using relative solvent-accessible surface area: buried$\leq$0.20; exposed$\geq$0.40 in the audited tasks.
S04 / S05	Local secondary structure (helix/strand/coil); global chain fold class (predominantly alpha helical, predominantly beta sheet, or mixed).
S06 / S07 / S08	Ligand-contact residues; metal-coordinating residues; the disulfide-bonded partner of a named cysteine. Frozen task-specific geometric rules determine the gold answer.
S09	Classify the $\chi1$ side-chain dihedral (N–CA–CB–CG), normalized to $[-180°,180°)$: g+ for $0\leq\chi1<120$; t for $\chi1\geq120$ or $\chi1<-120$; g− for $-120\leq\chi1<0$.
T01	Compare two supplied structural states under a fixed geometric question. No T01 task fits the selected RL cohort or this model's original-benchmark subset.

A residue is an amino-acid unit in a polymer chain; atom IDs refer to specific atoms within those residues. PDB is the coordinate-record format supplied to the model. Rotations and translations preserve distances and angles but change absolute coordinate values. The sequence-prediction tracks F01/F02 are separate benchmarks and are not mixed into any coordinate-reasoning headline here.

20. Original benchmark: all paired outcomes

The following table enumerates every eligible question. B=original baseline; RL=step 384; ✓=exactly correct, ×=incorrect. “Box” applies the same pre-existing terminal-box normalization to both outputs. Full prompts, raw responses, verifier details, and hashes are retained in the evidence bundle. Rows are ordered by frozen render ID, not by outcome.

Frozen render ID (representation / rotation suffix omitted)	B strict	RL strict	B box	RL box
G01-fb_7t9r-c7993bcf	×	✓	×	✓
G02-fb_7v3o-3098d134	×	×	×	×
G02-fb_8k3b-efebcbc0	×	×	×	×
G02-plastocyanin-dd41895f	×	×	×	×
G03-fb_7t9r-c27105b2	×	✓	×	✓
G04-v2_2dlb_1-2080f98f	×	×	×	×
G04-v2_2gf4_1-1be73820	×	×	×	×
G04-v2_4lws_1-769d0511	×	×	×	×
G04-v2_7des_1-a4e98648	×	×	×	×
I01-v2_2xvc_1_assembly1-0961c27a	×	×	×	×
N01-fb_8jdx-d5bcead5	×	×	×	×
P01-fb_8zmm-763fe080	✓	✓	✓	✓
P01-insulin-8edee5ea	✓	✓	✓	✓
P01-leucine_zipper-dc24b069	×	✓	✓	✓
P02-crambin-5f487424	✓	✓	✓	✓
P03-crambin-e64aa612	✓	✓	✓	✓
P03-fb_7t9r-8598ff4d	×	✓	✓	✓
S01-fb_7v3o-a9fd902f	×	×	×	×
S01-fb_8k3b-20152ce1	×	×	×	×
S01-plastocyanin-2ddad07b	×	×	×	×
S02-ph_ubiquitin_thr12-e7ce5a3c	×	×	×	×
S03-v2_2dlb_1-62aac8ba	×	✓	×	✓
S03-v2_3d7i_1-6157f11a	✓	×	✓	×
S03-v2_6e6q_1-50e54b33	×	×	×	×
S04-v2_3d7i_1-f325bcb4	×	×	×	×
S05-v2_1w6x_1-4355ea75	×	×	×	×
S05-v2_1zuy_1-8bd93875	×	×	×	×
S05-v2_4esr_1-993afc94	×	×	×	×
S05-v2_6r5m_1-882f6cf8	×	×	×	×

Frozen render ID (representation / rotation suffix omitted)	B strict	RL strict	B box	RL box
S06-v2_1isu_1-7c046afc	×	×	×	×
S06-v2_4r4l_1-8894036c	×	×	×	×
S06-v2_6l0w_1-dce0da3e	×	×	×	×
S07-fb_7v3o-24ba31c6	×	×	×	×
S07-fb_7yuj-8eb6fe47	×	×	×	×
S08-af_interleukin_8-4ebfb994	×	×	×	×
S08-crambin-db77c715	×	×	×	×
S08-fb_8kce-6a67ea87	×	×	×	×
S09-fb_7v3o-c1d5a19b	×	×	×	×
S09-fb_8aga-7ea4857a	×	✓	×	✓
S09-fb_8jdx-145d10b6	×	✓	✓	✓

21. Interpretation and limits

For the strict comparison, the source-group interval includes zero. The observed point estimate should therefore be treated as imprecise evidence of transfer on this small panel, not as a definitive positive improvement.

For the box-corrected comparison, the source-group interval includes zero. The observed point estimate should therefore be treated as imprecise evidence of transfer on this small panel, not as a definitive positive improvement.

The original-benchmark rerun provides a direct measurement of transfer for the validation-selected checkpoint: strict task-weighted change **+15.00 points**, and box-corrected change **+7.50 points**. The family-macro changes are +19.44 and +12.96 points respectively. These matched comparisons are the relevant evidence for this benchmark; the 61.42% native validation peak is not substituted for a benchmark result.

The strongest established result from the pilot is a reproducible improvement in measured task performance, including a separate grouped test. The interpretation is narrower than a claim that the model has learned general molecular reasoning. Answer-format adaptation and class-frequency learning explain a material part of the pilot gain, while some audited valid-answer transitions demonstrate content improvements. The original benchmark adds a useful transfer check, but its 40 eligible questions and tiny family denominators cannot establish broad mastery.

Selection on repeatedly observed validation makes the peak development score optimistic as an estimate of future performance. The original benchmark was not used to select the checkpoint, but only one trained checkpoint, one training seed, and one sampled response per benchmark question were evaluated. The benchmark’s 28 protein-group bootstrap is not a substitute for repetitions across decoding seeds or independent training runs.

Context filtering favors smaller structures and changes answer distributions. Forty of 88 benchmark questions fit; 48 remain unanswered under this model’s native limit. Excluded questions are not counted as incorrect in the conditional accuracy, and that accuracy must not be presented as full 88-question coverage. The learned model has not acquired a larger context window through this RL procedure.

The regression audit strongly supports an actual loss of policy quality and exposes an inadequately constrained effective update recipe. It does not yet distinguish optimizer dynamics from numerical/backend/weight-sync contributors. A successful independent evaluation of 384 supports retaining that checkpoint; it does not justify continuing indefinitely under the same unstable recipe or claim that a proposed corrective ablation has already worked.

Training remains in progress toward the previously authorized 876-update limit. A separate existing follow-up selects the final best validation checkpoint after completion and runs full standalone validation plus the 1,370-task paired test. Those future results are not included or anticipated here. This report’s benchmark selection remains the frozen step 384 comparison and should be versioned if later experiments are added.

22. Reproducibility and evidence index

The completed original-benchmark evidence bundle is **1,963,077 bytes** and contains all 80 raw API responses, frozen requests, checkpoint selection, export verification, source snapshot, and per-file hashes. Durable URI: s3://marin-us-east-02a/marin/checkpoints/bizon/snowball-pdbthink-epoch/2026.10.15.1/exports/analysis/original-benchmark-step384-30c8efb4553825ac.tar.gz. SHA256: 30c8efb4553825acb5ff956c6da3611bd1ad1f7d6092b2f1751901f21e14d91d.

Artifact	Pinned identity / location
Base model revision	cfc1d845dae89b067cdc7250d0164abefa5a69cf
Dataset release / revision	v1.3.0 / 3734406cb97b1702844319f9a5d860cbbf8fe660
Marin commit	44c95a7914864fdab8b74d597052e30cbc92b8cc
MarinSkyRL commit	5663f6129f53bb6aaa09549fa6cb423571503e94
Standalone benchmark vLLM commit	01911be34fac8715347962a8d791ca34a879e01e
Benchmark aarch64 vLLM wheel SHA256	55724f215fc85099047785098d5d8df4e46a9eb1baeb37c36471e78d15bdb813
Adapter SHA256	6fa472524ddea6e042905f87615f83a07fd5fee6339bd0d9f48cae5956fbc969
Original benchmark fingerprint	dee3923c6a5f4996a02a5e42f3cc1ac31ca15b249763470fe10916e7b65a1bfc
Native chat-template SHA256	414d8559d363292c811e624fe2db04d52647ca5bd89acc53e98bd4c4f371302a
Best-checkpoint export	s3://marin-us-east-02a/marin/checkpoints/bizon/snowball-pdbthink-epoch/2026.10.15.1/exports/global_step_384/policy
Export manifest SHA256	dfdb9e4204d648c323aad14d258b5b9c94c15e0befcd2925496829d4777ee8e9
Benchmark source snapshot	File-by-file SHA256 values in original-benchmark plan; preserves local source state as well as git HEAD
Full-history cutoff	2026-10-16T17:33:23.823875+00:00; last update 584

Repository: github.com/timodonnell/snowball-post-training. All experiment sources and compact evidence are under experiments/001-pdbthink-coordinate. The PDF is generated by build_result_report.py; original_benchmark.py freezes requests and scores completed responses; run_original_benchmark.py owns the inference servers; export_benchmark_checkpoint.py freezes validation selection and verifies native export.

Evidence files: pilot-validation-comparison.json; pilot-test-comparison.json; pilot-evidence.json; epoch-regression-audit.json; full-training-history.json; epoch-resume-resolved-launch.json; cohort_manifest.json; original-benchmark-comparison.json; original-benchmark-manifest.json. Figure sources: plot_pilot_evidence.py, plot_regression.py, plot_full_training.py, and plot_original_benchmark.py. Raw evidence is preserved separately with manifests and SHA256 checksums.

W&B histories: pilot xlyyl24u; first continuation ppc4ryud; resumed continuation e2s5w3qn. Native stored evaluation records are authoritative for checkpoint selection; external custom summary writes during training can be overwritten by the active W&B writer.

The regression evidence bundle is preserved at s3://marin-us-east-02a/marin/checkpoints/bizon/snowball-pdbthink-epoch/2026.10.15.1/exports/analysis/regression-through565-cb8cf1f493c6cbd7.tar (1,277,900,800 bytes; SHA256 cb8cf1f493c6cbd757e5b7dc6acdf19e9f6ccd2f1783c1e1c1a0bf5062a141b8). This report relies on saved experiment records and pinned implementation source, not on an unverified assumption about the current upstream defaults.

23. Exact validation membership identifiers

Full validation is the immutable validation.parquet in the prepared input artifact. Monitor membership is the immutable monitor.parquet subset of that same validation split. The complete per-task ID and prompt-hash membership is in cohort.json, with the checksum below. These identifiers distinguish a change in model/serving behavior from a change in the evaluated questions.

Frozen artifact	SHA256
cohort.json	92c85e37eb99d918f1ef4fe38f1b31f5b2434482f8386d048b70fe3bbb5d82c0
validation.parquet	5627d95cc0050f1b8de8add89e5bad3312d8398464914e46b6f46f6ce72eed8b
monitor.parquet	5e6223798c327228b145133333f8d52ab4f2a5c0e744e3e0e71a5434b4ada1fc
test.parquet	7a5ceb080eca609fa10d49931a1152443e851e41031bba7ebe474132b0f99a2d

Validation source-group ID	Tasks	Correct 384	Correct 512
group-0a598de775fce0cbbd92a1d5	27	23	11
group-1b872b3ca152b0c34dac8501	35	18	9
group-1ddf4c0d1f51a95b839313b3	110	61	41
group-2a905cb425c978eedbb1b046	41	28	12
group-42508844daba0ac859cefd40	41	27	9
group-4bab73dab4889558cfe72337	51	24	8
group-62e85cc5642bc4890807d115	38	20	7
group-95f23ea96073d1bba5c42389	36	22	11
group-9711b21ec18fd25e13e46350	38	19	8
group-9827b67dcbe2d8f9d7dbffe1	31	24	12
group-a03b21c590c51e217104e404	173	115	77
group-b2cd3b81804efe99ea2d851b	40	18	9
group-c0bb94b02503c173095621fc	92	75	41
group-c1ba54a50096f5b26cb7bc7a	42	15	9
group-cb7ddfceb8fd51cec5f33067	33	26	13
group-d4b34ad8f14bf85af570ec3e	38	19	12
group-dc953f8ed8611fbb049a5a45	35	18	5
group-f0657f60e5ddf1fc2684c28d	29	22	21
group-f9faf1231ce754aa477353f6	43	22	9
group-fa1894aecf11319f1f2abf5c	30	20	10

Durable input artifact: s3://marin-us-east-02a/marin/bizon/snowball-pdbthink/inputs/2026.10.07-v3. Neither the 1,003-task validation membership nor its 20 source groups changed between the long-run checkpoints. The only native cohort expansion was 107→1,003 tasks at the pilot/continuation boundary.

What did Snowball actually learn?

A paired rollout audit of validation at steps 0 and 384

16 October 2026 · 100 questions · 200 archived responses · 14 task categories

The improvement is mixed. There is concrete evidence of better coordinate retrieval, distance arithmetic and some salt-bridge calculations. But several higher-scoring categories are explained by class defaults or an easier geometric proxy. The rotamer responses now attempt dihedral calculations, yet their mathematical correctness is much weaker than their final-bin reward suggests.

The strongest shortcut results: step 384 says solvent-exposed in all 118 parseable S03 responses and coil in all 110 parseable S04 responses. In G03, 104 of 108 answers agree with the nearest-alpha-carbon proxy, although the task asks for minimum heavy-atom distance.

Rotamers: all 13 reviewed step-384 traces attempt numerical geometry; none is verified correct end to end. Seven still earn reward 1. A recurring missing factor changes angles but leaves all 116 validation gold bins unchanged when applied to the correct coordinates. This particular error is invisible to the category-only reward on these tasks.

This supplement qualifies the molecular-reasoning interpretation of the original result report; it does not change its measured scores. The matched original-benchmark result remains 5→11 of 40 strictly, or 8→11 after the existing box correction, with wide uncertainty. The present archival validation audit uses different inference paths at the two checkpoints and must not replace that matched comparison.

The accompanying HTML viewer contains every full prompt and response beside its review. The compressed JSON contains the same frozen sample for machine inspection. No new model inference or GPU job was required for this audit.

Sampling, scope and interpretation

I selected **100 question IDs before inspecting their reasoning**, then reviewed both archived answers: 100 step-0 rollouts and 100 step-384 rollouts. Pairing the questions lets us compare how each answer changes. Python random.Random uses seed **20261016**; the manifest was frozen at 2026-10-16T17:50:18.558924+00:00.

Sampling is random within strata but deliberately disproportionate. Each of the 14 present families receives seven questions, except G04, which has only three and is reviewed in full, and S09, which receives 13 because the user specifically requested rotamer analysis. Within a family, quota is allocated as evenly as possible across the available joint success states, subject to each stratum's capacity; ties use a seeded shuffle. IDs are sampled uniformly without replacement inside each stratum.

The strata are **00** (both fail), **01** (step-0 fail, step-384 succeed), **10** (step-0 succeed, step-384 fail), and **11** (both succeed). For example, S09 contributes three 00, three 01, three 10 and four 11 questions. Its sample therefore scores 7/13 at each checkpoint by construction, even though full-category accuracy increases. G02 includes all four step-384 successes. **Do not estimate population accuracy from this sample.** All reported population scores and controls use the full category.

The underlying validation cohort is the same frozen 1,003-question, 20-source-group set; the sample spans 17 groups. Questions sharing a structure are correlated. I01 (interface contacts), S06 (ligand contacts), S07 (metal coordination), S08 (disulfide partner) and T01 (two-state comparison) have no eligible validation questions and cannot be assessed here.

Step 0 means the original Snowball checkpoint before this project's additional RL, not an untrained model. Step 384 is the validation-selected checkpoint requested by the user. The original-model answers come from baseline-validation-v2 standalone serving; step-384 answers come from the native training-evaluator archive. Both use the same frozen prompts, temperature 0, one answer and the remaining 32,768-token context. Their serving/execution paths differ. Thus observed differences are descriptive evidence, not a clean causal decomposition into training versus runtime effects.

What was checked

The sampler checks question membership, exact baseline request messages, baseline output budgets and native prompt hashes. Replaying the frozen native verifier reproduces all 2,006 recorded rewards. It saves the 100 selected IDs, all response SHA256 hashes, input/native-archive hashes and full prompts. Strict success includes the native requirements for formatting, truncation and refusal; a semantically correct but unparseable answer can score zero.

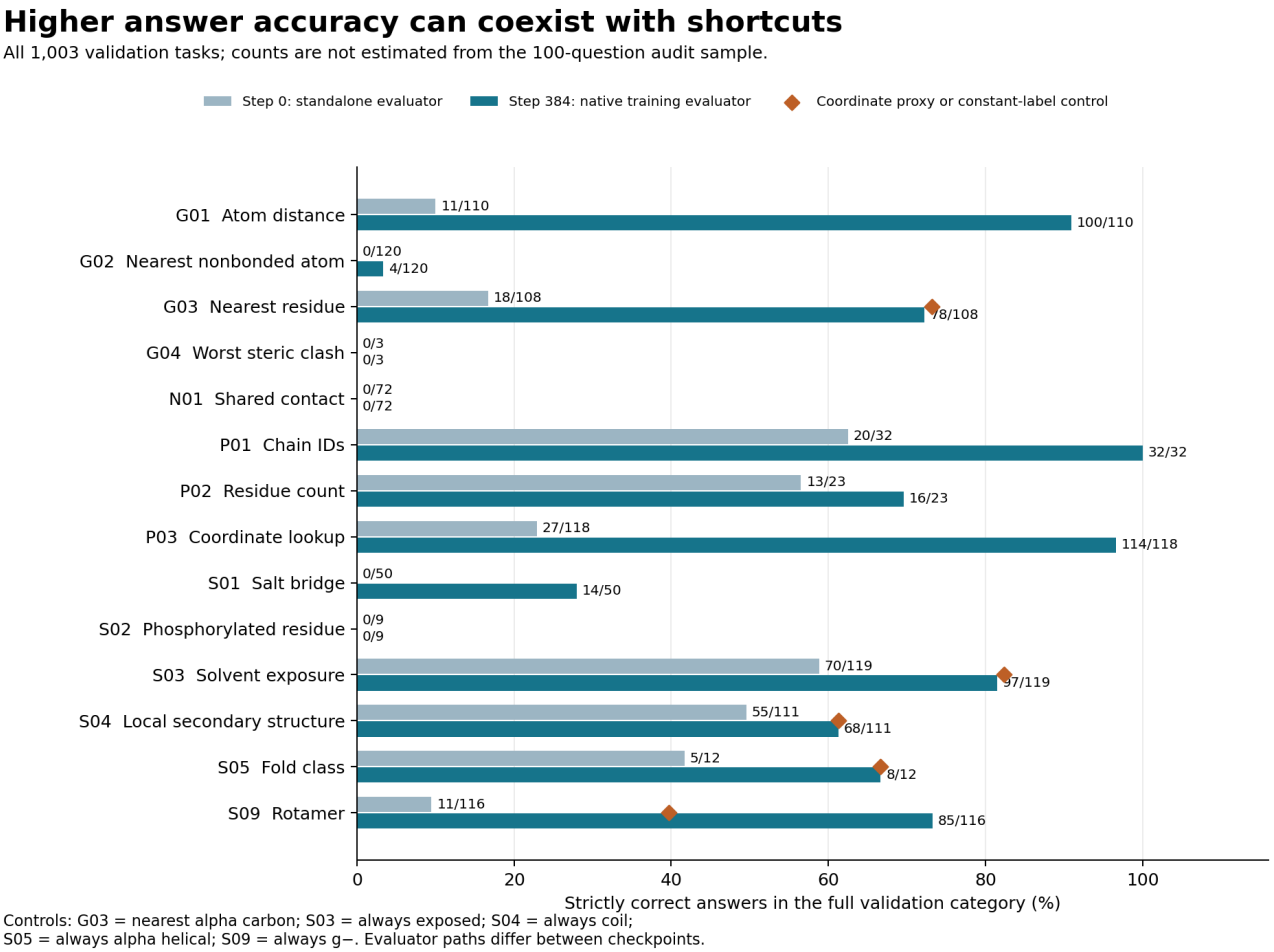
All 200 responses were inspected through text views that remove exact duplicate lines, show beginning/middle/end sections of long responses, and support targeted examination of computational passages. Repeated trailing invented metadata is retained in the raw artifact; the normal review view focuses on the substantive answer before the first JSON metadata block. Truncation and parsing failures were checked separately, including post-answer repetition. This was a qualitative Codex review, not a blinded independent-human or multiple-rater study. **Not every arithmetic line of every long trace was independently recomputed.**

Independent numerical checks parse fixed-column coordinates directly from the displayed prompts. They recompute all 116 S09 gold dihedrals using two equivalent formulas, all 108 G03 heavy-atom minima and their CA-only controls, all seven sampled G01 distances, all seven sampled P03 coordinate answers, and the true charged-atom contact for all seven sampled S01 questions. All recomputed gold answers agree. Unit checks cover known signed torsions, proper rotation/translation/scale invariance, a bin-changing counterexample to the wrong formula, and signed coordinate parsing.

The review distinguishes **correct final label**, **valid observable method**, **shortcut**, **format repair** and **unfinished generation**. A model’s prose is evidence of what it wrote, not proof of the internal mechanism that caused its answer. Claims about computation refer to visible operations; mentions of DSSP, Python or a database are not evidence that a tool ran. No tools were available to these model rollouts.

Family	Full N	Sample 00 / 01 / 10 / 11	Family	Full N	Sample 00 / 01 / 10 / 11
G01	110	2 / 2 / 0 / 3	P03	118	2 / 3 / 0 / 2
G02	120	3 / 4 / 0 / 0	S01	50	4 / 3 / 0 / 0
G03	108	2 / 2 / 1 / 2	S02	9	7 / 0 / 0 / 0
G04	3	3 / 0 / 0 / 0	S03	119	2 / 1 / 2 / 2
N01	72	7 / 0 / 0 / 0	S04	111	2 / 2 / 2 / 1
P01	32	0 / 4 / 0 / 3	S05	12	3 / 2 / 0 / 2
P02	23	2 / 2 / 2 / 1	S09	116	3 / 3 / 3 / 4

Full-cohort scores and simple controls



Controls: G03 = nearest alpha carbon; S03 = always exposed; S04 = always coil;
S05 = always alpha helical; S09 = always g−. Evaluator paths differ between checkpoints.

Bars: full-category strict accuracy at each archived checkpoint, with numerator/denominator labels. Diamonds: explicitly defined offline controls. These are conditional accuracies on context-eligible validation questions; the sample is used for qualitative interpretation only. The different inference paths prevent treating every bar difference as an isolated RL effect.

The controls require no learned molecular-reasoning policy: G03 selects the nearest CA; S03 always says exposed; S04 always says coil; S05 always says alpha helical; S09 always says g−. The S09 constant is clearly worse than the model. The S03/S04/S05 controls equal or exceed the model, so their score gains do not establish structural computation.

Category findings: P01 and P02

P01 · Chain identifiers

Full validation: **20/32 → 32/32**. Formatting and completion improve; the sampled chain-identification knowledge was already present.

Both checkpoints identify the right chain in all seven sampled questions. The four scored gains are output-contract repairs: boxed/inline FINAL is not parsed, or a trailing “Final Answer” overrides G with ANSWER. Step 384 gives a clean final field. These examples are single-chain structures.

How it fails: No failure remains at step 384 in the 32-question validation category. That does not establish more complex multichain parsing, nor does this sample show new molecular geometry knowledge.

P02 · Residue counting

Full validation: **13/23 → 16/23**. More consistent enumeration in some cases, with substantial counting and interpretation errors.

Case 41 recovers the 59-residue count after the baseline omits two positions. Case 42 enumerates 60 residues correctly where the baseline lists60 but claims83. Both checkpoints can count the 38-residue example in case 43.

How it fails: Case 39 regresses from 78 to 150 by mistaking the last author residue number for the count. Case 38 counts 22 then wrongly excludes TPO as a nonprotein triphosphate. Case 40 invents an extra entry. Case 44 gives the correct 29 but continues generating a repeated sequence until the length cap, so native reward is zero.

Category findings: P03 and G01

P03 · Coordinate lookup

Full validation: **27/118 → 114/118**. Strong evidence of improved retrieval and faithful copying of supplied coordinates.

The three sampled gains repair wrong coordinate content, including incorrect signs/components and wrong residue lookup. Case 49 changes x from 5.668 to the supplied−6.841; case 50 finds the requested R349 NH1 rather than another coordinate triple. The full category improves from 27/118 to 114/118.

How it fails: Lookup is still imperfect. Case 51 returns the SE atom of MSE1 instead of MSE33 at both checkpoints. Case 47 locates the correct residue context but substitutes a wrong CG2 triple. These tasks test reading the PDB, not deriving a physical quantity.

G01 · Atom distance

Full validation: **11/110 → 100/110**. Stronger atom lookup plus actual Euclidean distance calculation.

The model uses coordinate differences, squares, sums and a square root. This is substantive task-relevant behavior, already present in some baseline answers and substantially more reliable at step 384. In case 1, the baseline uses D73/L79 instead of D80/L94; step 384 selects the requested atoms and obtains23.86 Å. Case 3 now finishes the computation instead of exhausting its output budget.

How it fails: Residual failures include wrong residue/atom lookup, arithmetic and formatting. Case 6 substitutes a different oxygen and invents a serine CG. Case 2 correctly computes12.20 Å but never supplies a parseable FINAL field. Case 3 gives21.72 for a gold21.708 and passes the 0.02 Å tolerance despite a square-root error.

Category findings: G02 and G03

G02 · Nearest nonbonded atom

Full validation: **0/120 → 4/120**. Some coordinate-based search and bond-exclusion reasoning; no robust general solution.

All four step 384 successes in the entire120-question category are included because the success stratum is so small. They compute distances to alpha carbons and selected sidechain atoms, then sometimes recover the correct allowed atom in the target residue. Cases 10–13 show progress from sequence guesses, missing-target claims or unfinished loops.

How it fails: The searches are incomplete, and the covalent-bond graph is unreliable. Case 8 selects directly bonded GLU8 CB even though it must be excluded. Case 9 uses an incorrect arginine connectivity and selects CB instead of CG. Three-letter or otherwise invalid atom identifiers are common. Four successes out of 120 is weak evidence of general nearest-nonbonded-atom competence.

G03 · Nearest residue

Full validation: **18/108 → 78/108**. A clear coordinate shortcut: nearest alpha carbon substitutes for minimum heavy-atom distance.

All seven sampled step 384 traces explicitly compare CA–CA distances. Their answers all match the independently computed CA proxy. On the full108-question category,104 predictions match that proxy. The proxy itself scores 79/108; the model scores 78/108. This strongly supports a useful but incomplete geometric heuristic.

How it fails: The proxy is wrong on29 questions. The model follows its wrong answer on27 of those 29, and solves only one. In case 15, C192 is closer by CA distance (5.010 vs6.253 Å), but D191 is closer by the required heavy-atom minimum (2.976 vs3.298 Å). Cases 16 and18 show the same structural failure, with additional coordinate-copying mistakes.

Category findings: G04 and N01

G04 · Worst steric clash

Full validation: **0/3 → 0/3**. No demonstrated success at finding the worst steric clash.

The entire validation category has only three questions, and all three are reviewed. Step 384 makes more numerical attempts, including distance and radius arithmetic, but no response computes the required maximum positive atom-pair overlap reliably.

How it fails: It substitutes CA distances or residue-average radii, assumes unmeasured contact distances, mishandles bonded exclusions and omits relevant nonpolymer atoms. Cases 21 and23 report the least-negative overlap despite the requirement for a positive clash. Case 22 assumes2.8 Å contacts and averages residue radii. All three final identifiers are also invalid.

N01 · Shared contact residue

Full validation: **0/72 → 0/72**. More arithmetic, but the wrong contact definition and no demonstrated solution.

Baseline answers largely speculate about sequence midpoints or get stuck listing residues. All seven step 384 samples attempt CA-distance screens at the 4 Å cutoff, whereas the task requires any heavy-atom pair within 4 Å to each anchor.

How it fails: All seven trained responses truncate without a final answer; full-category performance remains0/72. A CA pair farther than 4 Å cannot rule out a sidechain contact. Misnumbered residues and repeated rescanning compound this invalid screening rule. Numerical effort alone is not evidence that the contact-intersection task was learned.

Category findings: S01 and S02

S01 · Salt-bridge partner

Full validation: **0/50 → 14/50**. Some real charged-atom distance calculations, especially for lysine partners.

Cases 54,56 and58 compute the specified oxygen–nitrogen distances and return the correct partners. Independent checks recover E860 OE2–K889 NZ=3.174 Å, E9 OE2–K13 NZ=2.863 Å and D6 OD1–K3 NZ=3.210 Å. Case 55 computes the correct R170–E162 contact but uses the wrong residue identifier format.

How it fails: Arginine searches often stop at NE or use CA/carboxylate-carbon proxies, missing NH1/NH2. Cases 52,53 and57 all have a qualifying NH1 contact even though every NE-to-partner-oxygen distance exceeds4 Å. Those step 384 traces truncate. This explains a concrete limit of the apparent salt-bridge procedure.

S02 · Phosphorylated residue

Full validation: **0/9 → 0/9**. No scored improvement, and several sampled answers lose correct biochemical identification.

The baseline often recognizes SEP as phosphoserine or TPO as phosphothreonine but fails to convert to the requested parent one-letter residue identifier, e.g. Y:S26. Step 384 still recognizes TPO in cases 62 and64, but omits the residue number.

How it fails: In cases 60,63 and65, step 384 instead calls SEP/TPO a separate small molecule, sulfate or nucleotide and guesses an ordinary Ser/Thr. Case 59 ignores phosphate atoms and guesses a common serine site. All seven sampled responses and all nine validation tasks fail. This category illustrates why a zero-to-zero score can hide semantic regressions.

Category findings: S03 and S04

S03 · Solvent exposure

Full validation: **70/119 → 97/119**. A near-constant solvent-exposed answer; no demonstrated solvent-accessibility calculation.

All seven sampled step 384 answers assume a typical accessible area or invent a relative SASA value, without computing occlusion from surrounding atoms. Across all 119 tasks,118 parsed answers are solvent-exposed and one is unparsed; none is buried.

How it fails: Always answering solvent-exposed scores 98/119, slightly above the model's97/119. All 21 buried questions fail. Cases 68 and69 regress from correct burial heuristics to exposure. The score increase is therefore not evidence that the model learned to calculate solvent accessibility.

S04 · Local secondary structure

Full validation: **55/111 → 68/111**. A near-constant coil answer; no demonstrated DSSP or backbone-geometry computation.

All seven sampled step 384 answers say coil. They refer to imagined DSSP runs, generic sequence propensities or invented phi/psi values. No actual tool is available or called. Across111 tasks,110 parsed predictions are coil and one is unparsed.

How it fails: Always answering coil scores 68/111, exactly the model's score. It solves none of the 33 helix or10 strand questions. Cases 73 and74 lose correct helix guesses. The apparent gain reflects class selection and output reliability, not demonstrated secondary-structure assignment.

Category findings: S05 and S09

S05 · Overall fold class

Full validation: **5/12 → 8/12**. An alpha-helical class preference and sequence-propensity rhetoric, without structural calculation.

The sampled responses do not calculate hydrogen bonds, backbone angles or helix/strand fractions. At step 384,10 of 12 full-category predictions are predominantly alpha helical and two are mixed. All eight correct answers are alpha-helical chains.

How it fails: Always answering alpha helical also scores 8/12. The single beta-sheet and all three mixed chains fail. Cases 81 and85 show wrong helix guesses; case 80 remains wrong mixed. One sampled gain (case 82) merely formats the baseline’s already correct helix guess.

S09 · Sidechain rotamer

Full validation: **11/116 → 85/116**. A more consistent explicit torsion-calculation template, but not a validated dihedral algorithm.

All 13 sampled step 384 responses attempt coordinate subtraction, cross products and angle calculation. Nine start with the correct four coordinates; four have clear extraction errors. Seven receive reward1, but none survives all checks of coordinates, formula, arithmetic and bin mapping. This is more than an always-g− rule, yet less than reliable rotamer computation.

How it fails: A recurring formula omits the middle-bond norm; there are also wrong atom selections, sign transcription errors, lost atan2 quadrants and positive-angle-to-g− bin mistakes. Correct category rewards frequently tolerate an incorrect angle. The dedicated rotamer section shows exactly why.

Rotamers: calculation or trick?

The answer is a mixture: **the visible step-384 behavior is an explicit calculation template, not simply a constant rotamer guess, but it is not a reliable dihedral algorithm.** It extracts N, CA, CB and a residue-specific fourth atom (e.g. CG or OG1), forms bond vectors and cross products, estimates an angle, and assigns one of three bins. The full category has 46 g−, 37 trans and 33 g+ labels; step 384 scores 85/116 versus 46/116 for always g−.

Let A=N, B=CA, C=CB and D=the fourth atom. Define u=B−A, v=C−B, w=D−C, n_0=u×v and n_1=v×w. With the benchmark's signed convention, a correct formula is:

χ_1 = atan2(‖v‖ (n_0 · w), n_0 · n_1), converted to degrees and wrapped to [−180°,180°).

A recurring trace formula is **atan2(n_0·w, n_0·n_1)**, missing the factor ‖v‖. The numerator and denominator then have different length dimensions, and changing coordinate scale changes the computed angle. This is not fixed by merely using atan2. Other traces subsequently replace terms, lose the quadrant, or make arithmetic errors.

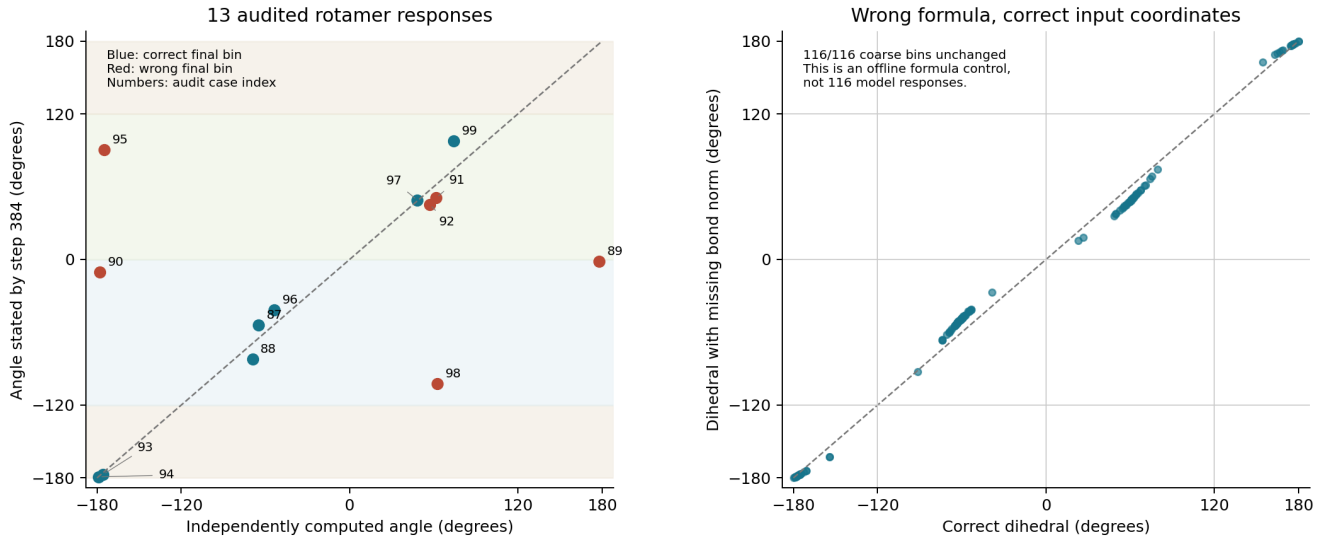
Yet applying this wrong formula to the correct four coordinates for every S09 validation question preserves **116/116 bins**. That is an offline formula control, not a claim that the model calculated all 116 angles correctly. The benchmark asks for broad bins: g− is −120°≤χ<0°, g+ is0°≤χ<120°, and trans is the rest. The observed geometries do not expose this formula error at those boundaries. A synthetic115° dihedral with a1.53-length middle bond crosses from g+ to trans under the wrong formula, so the shortcut is not generally valid.

In the 13 reviewed step 384 traces, four have clearly wrong input coordinates, two map a positive computed angle to g−, and one explicitly loses the atan2 quadrant. These counts are review findings in a biased sample, with overlapping error types. None of the 13 passes all stages of an end-to-end computation; nevertheless seven final bins are rewarded. At step 0, some correct bins likewise follow invented coordinates or a guess. Correctness of the baseline label is not a trustworthy procedural control either.

This is consistent with reward favoring a reusable, often successful template that does not need exact intermediate mathematics. It does not establish deliberate exploitation or prove that the written rationale causes the final answer. Stronger causal evidence would require controlled interventions on coordinates and comparison under matched inference settings.

Rotamer diagnostic plots

Rotamer reward does not certify the angle calculation



Bins: g− = [−120°, 0°); g+ = [0°, 120°); t = remaining angles. Correct bins can follow incorrect mathematics.
Left: outcome-stratified sample, not a population accuracy estimate; stated angles include transcription and arithmetic errors.

Left: independently recomputed angle versus the last numerical angle discussed in the sampled step384 trace; annotations are zero-based audit case indices. Blue means correct final bin, red wrong final bin. Angles are periodic, and the numbers reflect the model's own inconsistent calculations, not a clean implemented formula. Right: the isolated missing-norm formula on correct input coordinates for all116 questions. This separates a formula defect from input-copying and later arithmetic defects.

Case	True angle / bin	Step384 statement	Why the score is misleading or fails
87	−64.851° / g−	about−54° / g−	Right coordinates; missing norm; correct category.
89	+177.692° / t	−1.508° / g−	atan(y/x) used in place of atan2; quadrant lost.
90	−177.982° / t	about−10.3° / g−	Lys NZ substituted for backbone N.
91	+61.758° / g+	about+50.8° / g−	Positive bin mislabeled g− after coordinate arithmetic.
92	+57.067° / g+	about+45.14° / g−	Positive-angle sign/bin mistake, despite correct input points.
94	−178.782° / t	about−179.2° / t	Missing norm, but correct quadrant/wrapping and category.

Implications for the RL result

What is supported: the policy became much better at reading requested coordinates and completing simple distance calculations in this validation distribution. Some salt-bridge responses also use the right charged-atom geometry. Those are meaningful improvements. The separate matched original-benchmark result is still encouraging, although its uncertainty and limited coverage remain as reported.

What is not supported: a broad claim that the model learned reliable molecular structure analysis. Exposure and local secondary-structure gains are explained by constant labels. Fold classification reaches only the majority-class score. Nearest-residue success mostly follows a CA-only approximation. Rotamer categories often pass while the intermediate geometry is wrong. The hardest searches remain mostly or entirely unsolved.

These findings explain why aggregate validation accuracy is an incomplete progress measure, especially after the context filter changes class frequencies. They qualify the step 384 peak. They do **not** by themselves identify the cause of the later384→512 optimization regression, which requires the separate training/runtime audit in the main report.

A useful next experiment would keep the existing benchmark frozen and add a separate diagnostic set: balanced buried/exposed and helix/strand/coil labels; nearest-residue cases where CA and heavy-atom rankings disagree; rotamers near±120° and0°; atom-name distractors such as N versus NZ; and rigidly rotated copies plus chemically valid torsion changes. A continuous-angle diagnostic would reveal errors hidden by coarse bins. No such new model experiment is claimed in this report.

Selection and calibration matter: step 384 was chosen using validation, this sample is conditioned on outcomes, several questions share a structure, and inference paths differ. A stronger learning claim should reproduce the behavior on fresh source groups under matched serving, with repeated decoding where appropriate. The current audit provides concrete hypotheses and failure examples for that work, not an independent test-set estimate.

Reproducibility files: sample-manifest.json (seed, allocation, selected IDs and hashes), sample.json.gz (full prompts and200 raw responses), annotations.json and review-notes.json (all100 case reviews), statistics.json (full-category counts), geometry.json (independent checks), and rollout-audit.html (filterable paired viewer). Code: sample_reasoning_audit.py, analyze_reasoning_audit.py, plot_reasoning_audit.py and build_reasoning_audit.py. All paths are relative to this experiment; artifacts are in results/reasoning-audit/.

Appendix: every reviewed pair

Indices are stable, zero-based positions in the frozen sample. Outcome states use the native strict scores. Notes describe visible reasoning, including cases where the correct answer content is not accepted by the scorer. Full text and prompt are available under the same index in the HTML viewer.

Case 0 · G01 · state 11

pdbthink-g01-37da72df37f7fdc06014
Gold: 15.035

Step 0: Correct atom lookup and Euclidean method; mistaken square self-correction still within tolerance.

Step 384: Correct lookup and Euclidean calculation; final 15.04 agrees within tolerance.

Case 1 · G01 · state 01

pdbthink-g01-42d1e9b9cd0c41faec01
Gold: 23.86

Step 0: Uses D73 OD1 and L79 CB instead of requested D80/L94; computes 20.58 and malformed final.

Step 384: Uses requested atoms and computes 23.86 correctly.

Case 2 · G01 · state 00

pdbthink-g01-8c9718bbadeec9826f24
Gold: 12.2

Step 0: Wrong coordinates; computes 19.08 and boxed final not parsed.

Step 384: Correct coordinates and 12.20 distance, but says "We'll present FINAL" inline; no parsed FINAL.

Case 3 · G01 · state 01

pdbthink-g01-9b4bd938bcad18869011
Gold: 21.708

Step 0: Copies long atom lists and truncates before finishing.

Step 384: Correct atoms and squared sum; root arithmetic slightly wrong (21.72), accepted within 0.02 tolerance of 21.708.

Case 4 · G01 · state 11

pdbthink-g01-c90fc18d48008cadf422
Gold: 19.54

Step 0: Correct coordinate lookup and distance 19.54 after correcting a square.

Step 384: Correct coordinate lookup and distance 19.54; long repeated checking.

Case 5 · G01 · state 11

pdbthink-g01-d32a21518d4be1f2a28a
Gold: 7.363

Step 0: Correct atoms and Euclidean method; small arithmetic error, 7.36 accepted.

Step 384: Correct atoms, squared sum and distance 7.36.

Case 6 · G01 · state 00

pdbthink-g01-e410c580c153c1b78514
Gold: 15.703

Step 0: Wrong NZ/OG coordinates; computes 18.04 rather than 15.703.

Step 384: Correct NZ, initially sees correct OG but substitutes a different residue's coordinates and invents a serine CG; computes 15.07.

Case 7 · G02 · state 00

pdbthink-g02-0a13ea30c05bcb6f5882
Gold: "B:R347:CB"

Step 0: Sequence-neighbor speculation, excludes same residue too broadly; malformed B:ARG348:CA.

Step 384: Partial distance search, confuses sequence/residue names and selects nonexistent generic VAL CG; malformed answer.

Case 8 · G02 · state 00

pdbthink-g02-45fa8c56214f3ef68d5a
Gold: "E:P9:CD"

Step 0: Checks distances and correctly identifies PRO9 CD, but writes three-letter E:PRO9:CD.

Step 384: Computes many distances but selects directly bonded GLU8 CB, which must be excluded; three-letter identifier also fails.

Case 9 · G02 · state 00

pdbthink-g02-93369b0b6d0088581bfa
Gold: "B:R52:CG"

Step 0: Uses neighboring CA distances as proxy for target CZ; guesses PRO53 CA; malformed final.

Step 384: Wrong arginine connectivity excludes CG incorrectly; selects CB instead of CG, also malformed identifier.

Case 10 · G02 · state 01

pdbthink-g02-a05334aaddb277a3ac2e
Gold: "A:L7:CA"

Step 0: Incorrect leucine connectivity; guesses from sequence and truncates.

Step 384: Computes target-to-CA distances, correctly keeps own CA three bonds away; correct answer but no exhaustive all-atom proof.

Case 11 · G02 · state 01

pdbthink-g02-a7b76ffd50d590819814
Gold: "B:R349:CD"

Step 0: Repeated claims calculation impossible without tools; truncates.

Step 384: CA screening plus selected sidechain distances identifies own CD at about 2.828 A; correct answer, incomplete global search.

Case 12 · G02 · state 01

pdbthink-g02-e13b8e044eeff159245b
Gold: "A:D140:CA"

Step 0: Long listing, speculative sequence neighbors, confused exclusions, truncation.

Step 384: Target-to-CA screening and own-residue checks; excludes CB and picks CA at 3.629 A; correct answer without exhaustive atom search.

Case 13 · G02 · state 01

pdbthink-g02-eac05bcbd6cf3cac559d
Gold: "A:R893:CB"

Step 0: Enumerates atoms, falsely claims target missing, loops to truncation.

Step 384: Computes many CA and selected sidechain distances; correctly identifies own CB about 2.909 A, checks three-bond eligibility.

Case 14 · G03 · state 11

pdbthink-g03-aaa38eebbd31bc7e1b20
Gold: "A:V97"

Step 0: Chooses sequence-nearest V97, explicitly assumes chain linear; right label for invalid rationale.

Step 384: Uses CA-CA distances, V97 wins; correct label but not requested heavy-atom minimum.

Case 15 · G03 · state 00

pdbthink-g03-b8c52dcf813b5fdbaf68
Gold: "A:D191"

Step 0: Sequence-nearest C192 guess, boxed unparsed.

Step 384: Explicitly substitutes CA-CA distance and chooses C192; true all-heavy-atom winner D191.

Case 16 · G03 · state 00

pdbthink-g03-b8dd716f48aaca153aa8
Gold: "A:K254"

Step 0: Guesses K254 using sequence/sidechain arguments, even suggests Lys-Lys salt bridge; malformed final.

Step 384: Uses CA-CA proxy with miscopied coordinates; picks G255, also the true CA-proxy winner, instead of all-heavy-atom winner K254.

Case 17 · G03 · state 01

pdbthink-g03-b8e42ce58c960e7a9581
Gold: "A:V128"

Step 0: Sequence-nearest V128 guess is right but boxed unparsed.

Step 384: Numerical CA-CA comparison picks V128; correct label via proxy.

Case 18 · G03 · state 10

pdbthink-g03-d39788371b897d2c6854
Gold: "A:A726"

Step 0: Confused lookup and sequence guess picks A726 correctly.

Step 384: Uses CA-CA proxy (one coordinate is miscopied) and picks R727; true CA-proxy winner is also R727, but all-heavy-atom winner is A726.

Case 19 · G03 · state 11

pdbthink-g03-d3d17304af054d3a8090
Gold: "C:S15"

Step 0: Attempts selected atom distances and picks S15 correctly.

Step 384: Only CA-CA distances, picks S15 correctly.

Case 20 · G03 · state 01

pdbthink-g03-ff62f34e606e9930f42c
Gold: "B:R9"

Step 0: Misidentifies ASN as LYS and gets stuck listing atoms; truncates.

Step 384: Recognizes ASN and computes CA-CA distances, picks R9 correctly.

Case 21 · G04 · state 00

pdbthink-g04-55369d6376f773133850
Gold: "E:E8--E:P9"

Step 0: Attempts many distances but repeats identical estimates and truncates.

Step 384: Partial pair search finds no positive overlap, then reports least-negative TPO--MET; invalid criterion and identifier.

Case 22 · G04 · state 00

pdbthink-g04-ec33a251c9d7f1fddc18
Gold: "C:L122--C:T21"

Step 0: CA-distance shortcut with miscopied coordinates; picks TPO14--SER15, invalid identifier.

Step 384: Uses residue-average radii and assumed 2.8 A separation instead of closest atom pair; guesses LEU--GLU, invalid identifier.

Case 23 · G04 · state 00

pdbthink-g04-ed3c5d17f53366383bb7
Gold: "A:G1--A:L10"

Step 0: Excludes nonpolymer ligand, loops about needing tools and truncates.

Step 384: CA-CA surrogate and selected contacts; chooses least-negative overlap GLU9--ILE10, misses ligand clash and invalid identifier.

Case 24 · N01 · state 00

pdbthink-n01-042badfe4ccddfb8399e
Gold: "A:S33"

Step 0: Confused residue numbering and midpoint/sequence guesses, loops to truncation.

Step 384: Enumerates CA distances with 4 A cutoff instead of all-heavy-atom contacts; truncates without answer.

Case 25 · N01 · state 00

pdbthink-n01-0e296986440388362055
Gold: "A:D729"

Step 0: Midpoint heuristic guesses residue729 correctly but omits amino-acid code.

Step 384: CA-distance screen finds no shared contact and repeats calculations to truncation.

Case 26 · N01 · state 00

pdbthink-n01-6aba22c1ed9650543bde
Gold: "A:N34"

Step 0: Confused sequence/residue identity, midpoint speculation, truncation.

Step 384: CA-distance screen and even self-contact speculation, truncates without finding N34.

Case 27 · N01 · state 00

pdbthink-n01-6c054aa0e97241324843
Gold: "B:E66"

Step 0: Sequence-based guess GLU69, wrong partner.

Step 384: Misnumbers residues, computes CA distances, then repetitive verbal loop; truncation.

Case 28 · N01 · state 00

pdbthink-n01-9c4ded25b7879e22b8a0
Gold: "A:F696"

Step 0: Misidentifies CYS as MSE; midpoint guess696, missing residue code and refusal flag.

Step 384: CA-distance screening with confused identities, truncates without finding F696.

Case 29 · N01 · state 00

pdbthink-n01-dde84c4aa13319672701
Gold: "A:L257"

Step 0: Sequence-position/lysine prior guesses K258, boxed unparsed.

Step 384: Scans CA distances rather than atom contacts; truncates without finding L257.

Case 30 · N01 · state 00

pdbthink-n01-f3ee21661ab54b5f2428
Gold: "A:K5"

Step 0: Invents unbounded A:A residue list to thousands, truncates.

Step 384: Numerical CA-distance screening but wrong contact surrogate; truncates.

Case 31 · P01 · state 11

pdbthink-p01-1c5de69b97a988c5ff37
Gold: ["A"]

Step 0: Correctly reads single chain A.

Step 384: Correctly reads single chain A, concise FINAL.

Case 32 · P01 · state 11

pdbthink-p01-3626b2c2edd92266bd9e
Gold: ["B"]

Step 0: Correctly reads single chain B.

Step 384: Correctly reads single chain B, concise FINAL.

Case 33 · P01 · state 01

pdbthink-p01-4e6d164114d60bc16ef6
Gold: ["Y"]

Step 0: Correctly reads chain Y, but FINAL only inline/boxed and unparsed.

Step 384: Same correct chain Y in valid FINAL format.

Case 34 · P01 · state 01

pdbthink-p01-886b952a14d72529f9e7
Gold: ["E"]

Step 0: Correctly reads chain E, but FINAL only inline/boxed and unparsed.

Step 384: Same correct chain E in valid FINAL format.

Case 35 · P01 · state 01

pdbthink-p01-b04a7082daaf0d04c2d7
Gold: ["G"]

Step 0: Correct chain G, but trailing "Final Answer" overrides parsed value with ANSWER.

Step 384: Correct chain G with clean final field.

Case 36 · P01 · state 11

pdbthink-p01-da81c82017f235e96bd8
Gold: ["A"]

Step 0: Correctly reads single chain A.

Step 384: Correctly reads single chain A, concise FINAL.

Case 37 · P01 · state 01

pdbthink-p01-f1b3aaf8b39d4ea2727d
Gold: ["C"]

Step 0: Correctly reads chain C, but FINAL only inline/boxed and unparsed.

Step 384: Same correct chain C in valid FINAL format.

Case 38 · P02 · state 00

pdbthink-p02-0e516812e3e00d5bb293
Gold: 22

Step 0: Lists22 residues but miscounts21.

Step 384: Counts 22 then wrongly excludes TPO as non-amino-acid triphosphate; answers21.

Case 39 · P02 · state 10

pdbthink-p02-3ba49f7068c2ac316262
Gold: 78

Step 0: Correct inclusive span150-73+1=78.

Step 384: Enumerates using author numbers73..150 then confuses last author number with count; answers150.

Case 40 · P02 · state 00

pdbthink-p02-3fc51a2a8e831be302f1
Gold: 35

Step 0: Lists35 distinct numbers, correct 35 but malformed final.

Step 384: Invents/duplicates sequence entries and residue368; answers36.

Case 41 · P02 · state 01

pdbthink-p02-535fc17ac77638245254
Gold: 59

Step 0: Omits residues12 and32 and returns57.

Step 384: Enumerates 59 positions and answers59; some sequence transcription remains inconsistent.

Case 42 · P02 · state 01

pdbthink-p02-556576bc66cde9224992
Gold: 60

Step 0: Enumerates 60 entries yet claims83; first residue also wrong.

Step 384: Correctly enumerates 224..283 as60.

Case 43 · P02 · state 11

pdbthink-p02-6837b5c90341cbb2aeed
Gold: 38

Step 0: Correctly enumerates 38 residues.

Step 384: Correctly enumerates 38 residues.

Case 44 · P02 · state 10

pdbthink-p02-dbad3107f376709840b0
Gold: 29

Step 0: Correct 29, excludes ligand.

Step 384: Correct 29 and valid FINAL, but generation continues and hits length limit; native truncation makes reward0. The raw tail is a long repeated sequence string after the correct answer.

Case 45 · P03 · state 11

pdbthink-p03-0deb94b2b5395dea3aff
Gold: [-4.689, 1.288, 6.099]

Step 0: Finds requested atom and copies correct coordinates.

Step 384: Finds requested atom and copies correct coordinates.

Case 46 · P03 · state 11

pdbthink-p03-1005fd0d9a9a7e260de1
Gold: [2.366, 1.446, 1.744]

Step 0: Finds requested atom and copies correct coordinates.

Step 384: Finds requested atom and copies correct coordinates.

Case 47 · P03 · state 00

pdbthink-p03-78deef03b65a3ee5debc
Gold: [1.185, 2.082, 3.636]

Step 0: Reports wrong coordinate triple for I189 CG2.

Step 384: Correct residue context but substitutes another coordinate triple for CG2; wrong answer.

Case 48 · P03 · state 01

pdbthink-p03-d6e1a628b1a14a793f0f
Gold: [0.524, -12.107, -3.51]

Step 0: Correct x but wrong y/z for Q32 CD.

Step 384: Correct full Q32 CD triple.

Case 49 · P03 · state 01

pdbthink-p03-dcf450aa37957ad9e600
Gold: [-6.841, -6.857, 3.708]

Step 0: Wrong x (5.668 rather than -6.841) for I724 CD1.

Step 384: Correct full I724 CD1 triple.

Case 50 · P03 · state 01

pdbthink-p03-e325d678f44432b07f87
Gold: [-7.831, -21.44, -1.959]

Step 0: Wrong NH1 coordinates and malformed final.

Step 384: Correct R349 NH1 coordinates and format.

Case 51 · P03 · state 00

pdbthink-p03-f4f1255c38a7abc2b4f3
Gold: [3.394, 4.922, -1.942]

Step 0: Returns SE of MSE1 rather than MSE33.

Step 384: Same MSE1-for-MSE33 error, now correctly formatted.

Case 52 · S01 · state 00

pdbthink-s01-14cf6f33b6243fb40ff2
Gold: "A:D151"

Step 0: Wrong/mixed coordinates; concludes GLU125 and encodes glutamate as G.

Step 384: Attempts acidic partners but CA proxies and incomplete nitrogen search; never finishes. Independent check: NH1-OD2=3.764 A satisfies contact, whereas both NE-OD distances exceed4 A.

Case 53 · S01 · state 00

pdbthink-s01-1898ea7f0fb5738b95c3
Gold: "A:D725"

Step 0: Sequence-based acidic-partner guess ASP685; malformed final.

Step 384: Calculates CA and then NE distances, confused acidic identities, truncates before complete nitrogen search. Independent check: NH1-OD1=3.768 A satisfies contact, whereas both NE-OD distances exceed6 A.

Case 54 · S01 · state 01

pdbthink-s01-387af46b559a66878f81
Gold: "A:K889"

Step 0: Sequence-nearest basic residue guess K876; malformed identifier.

Step 384: Uses explicit E860 OE1/OE2 to K889 NZ distances3.799/3.174 A, correct partner.

Case 55 · S01 · state 00

pdbthink-s01-a2888039258cd7aa6a8f
Gold: "A:E162"

Step 0: Sequence-neighbor speculation picks GLU162, but malformed identifier.

Step 384: Computes NE-to-OE1/OE2 distances3.745/2.696 A and finds correct partner, but writes A:Glu162 rather than A:E162.

Case 56 · S01 · state 01

pdbthink-s01-b0f0c59d7ce76313a097
Gold: "A:K13"

Step 0: Sequence/helix heuristic guesses Lys13 correctly, malformed identifier.

Step 384: Checks charged-atom distances and finds Lys13 NZ within 4 A of Glu9 OE2, correct one-letter identifier.

Case 57 · S01 · state 00

pdbthink-s01-f33d19b7091e744c8119
Gold: "A:D127"

Step 0: Atom/residue-number confusion and endless listing, truncates.

Step 384: Mostly uses Arg NE and carboxylate carbon/oxygen proxies, misidentifies residues and truncates. Independent check: NH1-OD1=2.999 A satisfies contact; NE-OD distances are4.375/4.756 A.

Case 58 · S01 · state 01

pdbthink-s01-fa708c8afeed9655d2c8
Gold: "B:K3"

Step 0: Starts coordinate lookup then unbounded invented sequence repetition, truncates.

Step 384: Checks Lys/Arg nitrogen to Asp6 oxygen distances and identifies Lys3; corrects chainA toB before final.

Case 59 · S02 · state 00

pdbthink-s02-0e901babb294a47b1aed
Gold: "X:S5"

Step 0: Incorrectly calls SEP selenocysteine, loops to truncation.

Step 384: Rejects SEP as nonprotein/triphosphate despite phosphate atoms, picks generic Ser by frequency prior; invalid identifier.

Case 60 · S02 · state 00

pdbthink-s02-4944f8404e8565535336
Gold: "B:S1493"

Step 0: Correctly identifies SEP1493 as phosphoserine but omits number in final.

Step 384: Misidentifies SEP as sodium ethyl phosphate and chooses terminal SER; invalid identifier.

Case 61 · S02 · state 00

pdbthink-s02-733ea89e853501b52c10
Gold: "B:S1212"

Step 0: Rejects SEP as phosphorylatable, speculates about other S/T/Y residues, truncates.

Step 384: Selects ordinary Ser1184/sequence-position14 from priors; invalid S:14 identifier.

Case 62 · S02 · state 00

pdbthink-s02-7f11b16bfe86ea3f49f3
Gold: "E:T5"

Step 0: Correctly identifies TPO5 as phosphothreonine but uses three-letter modified code.

Step 384: Eventually identifies TPO but omits number and required T one-letter code.

Case 63 · S02 · state 00

pdbthink-s02-b793e33d783844f71cb6
Gold: "Y:S26"

Step 0: Correctly identifies SEP26 as phosphoserine, but uses SEP instead of S.

Step 384: Wrongly calls SEP sulfate/nonprotein and infers Thr25 phosphorylation; invalid identifier.

Case 64 · S02 · state 00

pdbthink-s02-e98afd677b93afdd2b37
Gold: "E:T5"

Step 0: Correct TPO5 identification, wrong modified residue code.

Step 384: Recognizes TPO5 phosphothreonine but outputs E:TPO without number.

Case 65 · S02 · state 00

pdbthink-s02-f6e1880304ac9eb7dec6
Gold: "G:T378"

Step 0: Correctly identifies TPO378 but malformed identifier.

Step 384: Wrongly calls TPO phosphorylated thymidine/nucleotide, picks ordinary SER385 from frequency prior.

Case 66 · S03 · state 00

pdbthink-s03-1a87d8bc274e61c8c679
Gold: "buried"

Step 0: Default solvent-exposed guess, wrong buried Arg.

Step 384: Invents typical Arg RSA around0.45, no coordinate occlusion computation; wrong solvent-exposed.

Case 67 · S03 · state 01

pdbthink-s03-1eade67d68f9e20a8837
Gold: "solvent-exposed"

Step 0: Falsely claims P58 absent and loops/refuses to truncation.

Step 384: Invents typical Pro RSA0.45/0.46, no actual surface calculation; correct exposed label.

Case 68 · S03 · state 10

pdbthink-s03-375152b0490a9f864d2d
Gold: "buried"

Step 0: Misidentifies Glu as Asp; interior-position heuristic gives correct buried label.

Step 384: Uses charged-residue prior and invented typical RSA0.52; wrong exposed label.

Case 69 · S03 · state 10

pdbthink-s03-66571901f19bad228378
Gold: "buried"

Step 0: Hydrophobic Ile burial heuristic gives correct label.

Step 384: Assumes typical Ile environment is exposed, no computed SASA; wrong label.

Case 70 · S03 · state 11

pdbthink-s03-76c290cc180df4c0bb0e
Gold: "solvent-exposed"

Step 0: Polar/linear-chain heuristic gives correct exposed label.

Step 384: Invented typical GLN RSA0.45-0.55, no actual geometry; correct exposed label.

Case 71 · S03 · state 00

pdbthink-s03-93d1d33b8b0f9b3d7612
Gold: "buried"

Step 0: Gly/N-terminal exposure heuristic, wrong label and malformed final.

Step 384: Assumes Gly SASA near maximum, gives wrong exposed label.

Case 72 · S03 · state 11

pdbthink-s03-aee73415e190766108d4
Gold: "solvent-exposed"

Step 0: C-terminal Pro exposure guess gives correct label.

Step 384: Invented typical Pro SASA values, mentions Shrake-Rupley without executing it; correct exposed label.

Case 73 · S04 · state 10

pdbthink-s04-064ef4ff0d977acbdbba
Gold: "helix"

Step 0: Sequence helix-propensity guess happens to be correct.

Step 384: Claims simulated DSSP gives coil without calculations; wrong label.

Case 74 · S04 · state 10

pdbthink-s04-1113ec03b05888fda20c
Gold: "helix"

Step 0: Leucine helix-propensity guess happens to be correct.

Step 384: Misidentifies Leu145 as Glu and invents typical phi/psi values; wrong coil.

Case 75 · S04 · state 01

pdbthink-s04-4c7551ab5335ff00be01
Gold: "coil"

Step 0: Coil guess based on absent annotations, boxed unparsed.

Step 384: Conceptual Python comment asserts coil, no DSSP/angle calculation; right label and format.

Case 76 · S04 · state 01

pdbthink-s04-80cb05fdaeadb52db361
Gold: "coil"

Step 0: Leucine helix guess, malformed final.

Step 384: Repeated simulated-DSSP claims and coil prior, right label.

Case 77 · S04 · state 00

pdbthink-s04-ceb6ea12c11f39a9614a
Gold: "helix"

Step 0: Default coil in absence of annotations, wrong helix.

Step 384: Claims typical DSSP/known result coil, no actual calculation; wrong helix.

Case 78 · S04 · state 11

pdbthink-s04-d19843e976ccbfbe45c2
Gold: "coil"

Step 0: Defaults coil despite confusion over whether target exists; correct label.

Step 384: Invented typical phi/psi and inconsistent structural ranges, final coil correct.

Case 79 · S04 · state 00

pdbthink-s04-d973760836233043d5af
Gold: "helix"

Step 0: Polar-residue/loop propensity guess coil, wrong helix.

Step 384: Claims absent 3D data and imagined DSSP result coil, wrong helix.

Case 80 · S05 · state 00

pdbthink-s05-1da4447909b792b66e68
Gold: "predominantly alpha helical"

Step 0: Defaults mixed alpha/beta from chain length/lack of annotations; wrong helix.

Step 384: Imagined DSSP and generic composition prior, chooses mixed despite briefly saying helix; wrong.

Case 81 · S05 · state 00

pdbthink-s05-2b02ff803542cc679d69
Gold: "predominantly beta sheet"

Step 0: Default mixed in absence of annotations, wrong beta.

Step 384: Helix propensity/imagined DSSP claims, wrong beta.

Case 82 · S05 · state 01

pdbthink-s05-6491a9f0ba5f8398cf12
Gold: "predominantly alpha helical"

Step 0: Sequence helix-propensity guess correct, boxed unparsed.

Step 384: Same general helix-prior reasoning, no geometry, correct final format.

Case 83 · S05 · state 11

pdbthink-s05-935fd4833f2d5d88ac61
Gold: "predominantly alpha helical"

Step 0: Helix assumption without calculated angles or H bonds; correct label.

Step 384: Long confused atom/residue enumeration then generic helix prior; correct label.

Case 84 · S05 · state 11

pdbthink-s05-9922e3fbacc94f0967f1
Gold: "predominantly alpha helical"

Step 0: Invents long sequence span and helix rationale, correct label.

Step 384: Imagined DSSP and composition prior, no calculated structure, correct helix.

Case 85 · S05 · state 00

pdbthink-s05-ba17ccb48c5c29215a24
Gold: "mixed alpha/beta"

Step 0: Hesitant mixed guess is semantically correct but refusal/unparsed.

Step 384: Generic helix frequency prior changes answer to wrong helix.

Case 86 · S05 · state 01

pdbthink-s05-f90567232a33288834f5
Gold: "predominantly alpha helical"

Step 0: Unbounded repeated invented sequence, truncation.

Step 384: General helix-propensity argument, no geometry, correct helix and completion.

Case 87 · S09 · state 01

pdbthink-s09-00e61ebcad77f449d5ac
Gold: "g-"

Step 0: Correct input points but wrong torsion denominator (normal dot middle bond, zero by construction); -90 degrees gives right g-, boxed FINAL unparsed.

Step 384: Correct four points; atan2 omits middle-bond norm; -54 rather than -64.851 degrees, still g-.

Case 88 · S09 · state 01

pdbthink-s09-05ad620dea98eba272f9
Gold: "g-"

Step 0: Invented input coordinates and wrong norm-product denominator; -16.6 degrees and g- unparsed.

Step 384: N x sign copied incorrectly; missing-norm formula and inaccurate atan estimate yield about -82 degrees; g- correct despite wrong geometry.

Case 89 · S09 · state 10

pdbthink-s09-1238597964105a6e0234
Gold: "t"

Step 0: Invented coordinates and chemical misidentification; approximately -178.55 degrees yields correct t.

Step 384: Correct points; omits norm and replaces atan2 with atan(y/x), losing quadrant: -1.508 degrees instead of +177.692; wrong g-.

Case 90 · S09 · state 10

pdbthink-s09-18bb0a0277996b7c14dc
Gold: "t"

Step 0: Confuses NZ with N, later invents a different coordinate set; 169.5 degrees yields correct t.

Step 384: Uses terminal lysine NZ instead of backbone N; missing-norm calculation gives -10.3 degrees and wrong g- (true -177.982/t).

Case 91 · S09 · state 00

pdbthink-s09-27908dbdd70df9caa959
Gold: "g+"

Step 0: Misidentifies residue and later uses wrong coordinates; -34.6 degrees and malformed g-.

Step 384: Initially correct points, later CB-z transcription error; about +50.8 degrees but labels positive interval g-; true +61.758/g+.

Case 92 · S09 · state 00

pdbthink-s09-65f1805f38c56084285b
Gold: "g+"

Step 0: Unbounded atom enumeration, no finished angle or FINAL.

Step 384: Correct points; missing-norm angle +45.14 degrees, falsely claims prompt duplicates g- bin, answers g-; true +57.067/g+.

Case 93 · S09 · state 11

pdbthink-s09-6ea578199e0816dc8e23
Gold: "t"

Step 0: Invented coordinate set, inconsistent formulas; -170.9 degrees yields correct t.

Step 384: Correct points; missing-norm -177.326 degrees versus true -175.917; handles quadrant and t correctly.

Case 94 · S09 · state 01

pdbthink-s09-714945209b1ba67d620c
Gold: "t"

Step 0: Invented coordinate set and wrong formula; 103.5 degrees/g+, scoring also marks refusal.

Step 384: Correct points; missing-norm -179.2 degrees versus true -178.782; correct wrapping and t.

Case 95 · S09 · state 10

pdbthink-s09-8100dc4b872225eab41e
Gold: "t"

Step 0: Invented points include implausibly long bond; wrong formula still gives -177 degrees and correct t.

Step 384: Wrong N coordinates, other three correct; missing-norm 90.6 degrees/g+ instead of true -174.724/t.

Case 96 · S09 · state 11

pdbthink-s09-b432ff6735b8a390c2d6
Gold: "g-"

Step 0: Correct four coordinates but cross-product component and sign errors; eventually chooses correct g-.

Step 384: Correct points; missing-norm -41.6 degrees instead of -53.650, correct g-.

Case 97 · S09 · state 11

pdbthink-s09-c2f5987840cc6e243a68
Gold: "g+"

Step 0: Wrong N/CA coordinates and wrong norm-product formula; 38.3 degrees gives correct g+.

Step 384: Correct points but missing norm plus cross/dot/arctangent arithmetic errors; switches from 56.5 to 48.9 degrees, right g+ (true 48.318).

Case 98 · S09 · state 00

pdbthink-s09-df09cf8da6916725f250
Gold: "g+"

Step 0: Hallucinates coincident N/CA, repeats undefined-torsion argument and truncates.

Step 384: Drops x signs and mixes/fabricates N/CA components; -102.5 degrees/g- instead of true +62.575/g+.

Case 99 · S09 · state 11

pdbthink-s09-e07a5c69968b5c356336
Gold: "g+"

Step 0: Explicit educated rotamer-prior guess with invented coordinates; no completed valid calculation, correct g+.

Step 384: Correct initial points; starts missing-norm formula, then substitutes unrelated cross-product component into denominator; 97.5 degrees gives correct g+ (true 74.291).